Artificial Intelligence
UNIT I
Introduction

Prepared By

Mrs. Priyadarshani Doke

Syllabus

- Introduction to Artificial Intelligence
- Foundations of Artificial Intelligence
- History of Artificial Intelligence
- Limits of AI
- Ethics of AI
- Future of AI
- AI Components
- AI Architectures.
- Intelligent Agents
  - Agents and Environments
  - Good Behavior: Concept of Rationality
  - Nature of Environments
  - Structure of Agents

- **Case Study:** Autonomous Taxi Agent – Waymo One, AI in Healthcare – IBM Watson for

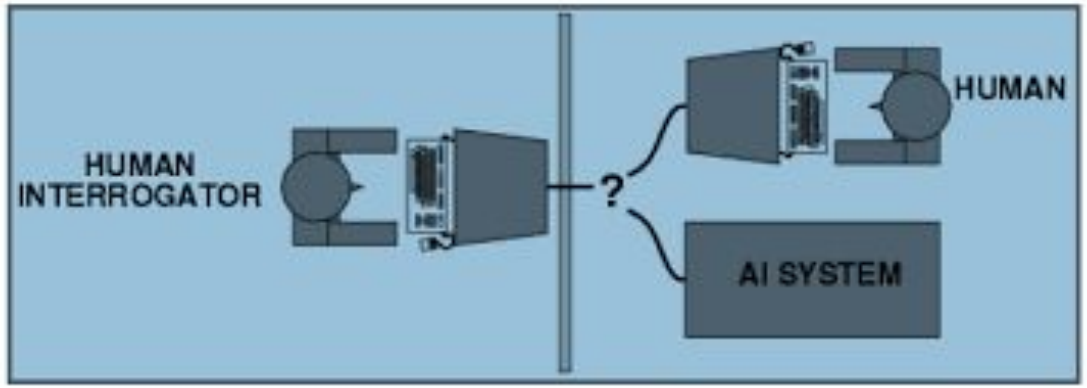
Introduction to Artificial Intelligence

- What is AI:

	Success in terms of fidelity to *human* performance ↓	success in terms of *ideal* performance called **rationality** ↓
thought processes and *reasoning* →	**Thinking Humanly *(System that think like human)*** "The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	**Thinking Rationally *(System that think rationally)*** "The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Behavior →	**Acting Humanly *(System that act like human)*** "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	**Acting Rationally *(System that act rationally)*** "Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Introduction to Artificial Intelligence

- What is AI:
  - Acting humanly: Turing Test Approach
    - Turing (1950) "Computing machinery and intelligence":
    - Operational test for intelligent behavior: the Imitation Game
    - Interrogator asks questions of two “people” who are out of sight
    - 30 minutes to ask whatever he or she wants
    - Task: to determine only through the questions and answers which is which (human or AI system)
    - Computer deemed intelligent if interrogator can’t distinguish between person and computer.
    - Artificial intelligence is the enterprise of constructing an artificate that can pass the Turing text



Introduction to Artificial Intelligence

Acting humanly: Turing Test Approach

- What major components were important
  - Natural language processing
    - to enable it to communicate successfully in English
  - Knowledge representation
    - to store what it knows or hears;
  - Automated reasoning:
    - to use the stored information to answer questions and to draw
  - Machine learning:
    - to adapt to new circumstances and to detect and extrapolate patterns
- What additional for total Turing Test
  - Computer vision
  - Robotics
- Note: looking at I/O behavior only

Introduction to Artificial Intelligence

- **Thinking humanly**: cognitive modeling Approach
  - **Cognitive Science** – must figure out how human's think
    - To say program thinks like human need to find actual working of human
    - Three ways to find actual working of human mind
      - **Through introspection**—trying to catch our own thoughts as they go by
      - **Through psychological experiments -** observing a person in action
      - **Through brain imaging** —observing the brain in action
    - How to validate? Requires
      - Predicting and testing behavior of human subjects (top-down)
      - Direct identification from neurological data (bottom- up)
      - Requires scientific theories of internal activities of the brain

Introduction to Artificial Intelligence

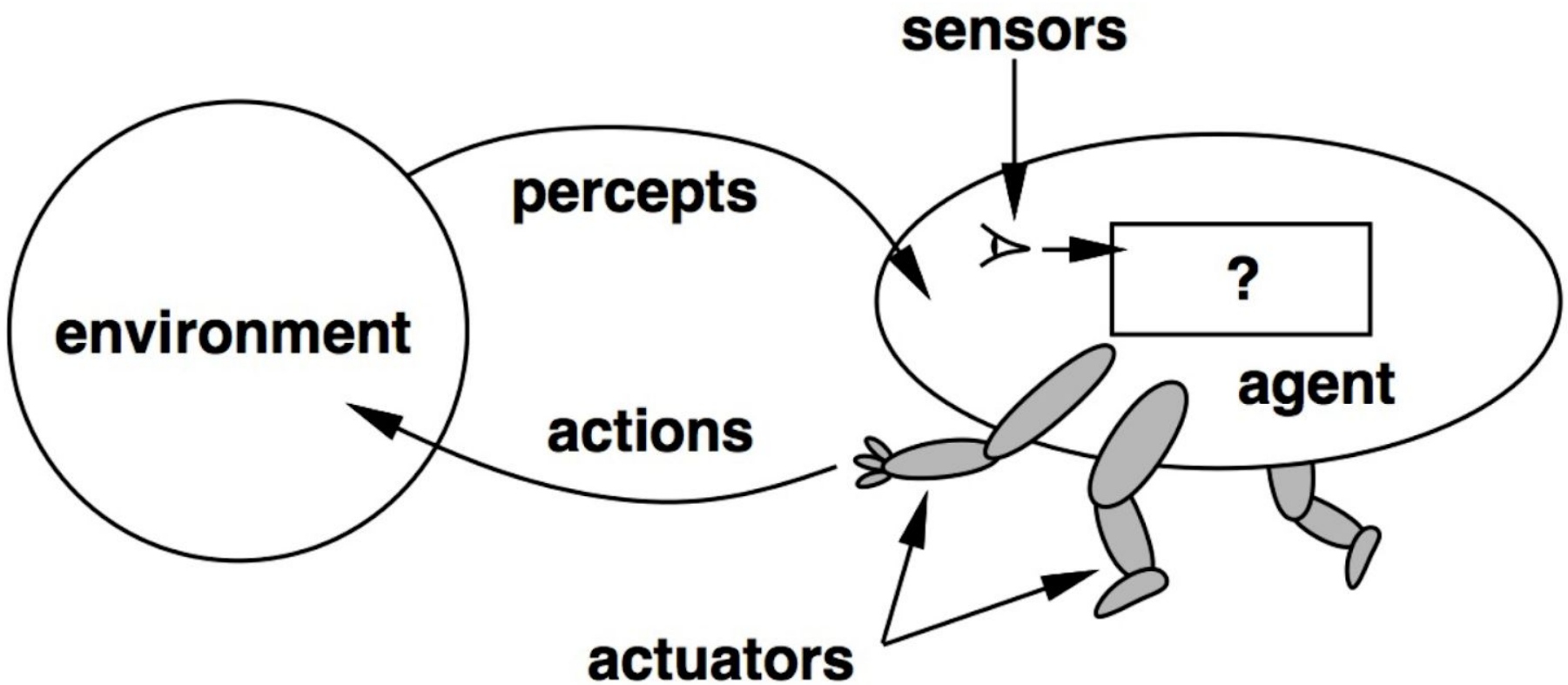
- **Thinking rationally: "laws of thought"** Approach
  - **Aristotle**: what are correct arguments/thought processes?
  - The ability to think with reason
  - It encompasses the ability to draw sensible conclusions from facts, logic and data. In simple words, if your thoughts are based on facts and not emotions, it is called rational thinking.
  - Several Greek schools developed various forms of logic:
    - notation and rules of derivation for thoughts;
    - may or may not have proceeded to the idea of mechanization
  - Direct line through mathematics and philosophy to modern AI
  - Problems:
    - Not all intelligent behavior is mediated by logical deliberation
    - Some knowledge is very hard to encode – informal, uncertain
    - In practice, computationally intractable

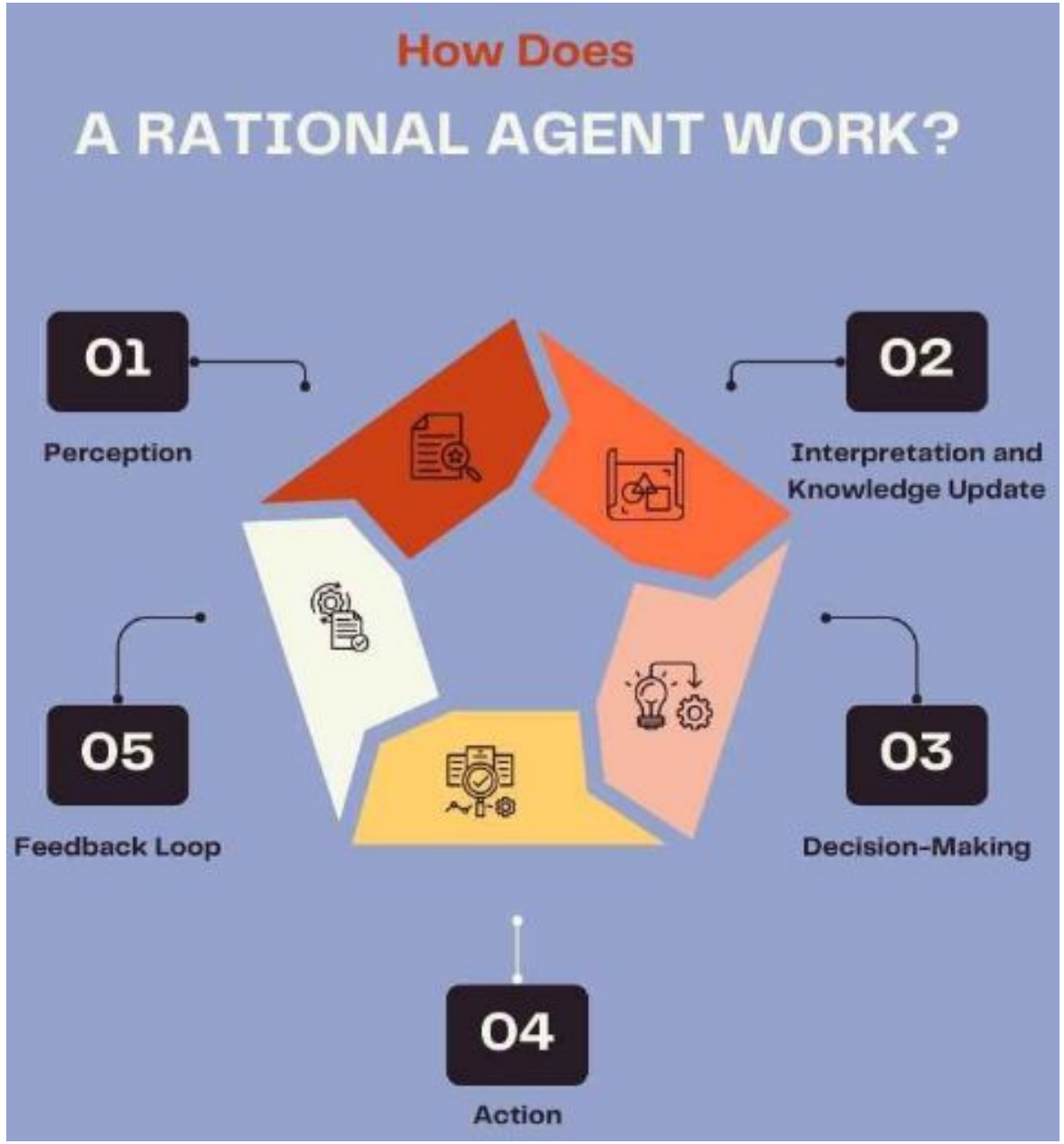
Introduction to Artificial Intelligence

- **Acting rationally**: rational agent Approach
  - An **agent** is just something that acts
  - computer agents are expected to do more:
    - Operate autonomously,
    - perceive their environment,
    - persist over a prolonged time period,
    - adapt to change,
    - create and pursue goals

Rational agent

- A **Rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- Correct thinking is good but:
  - Sometimes you must do something and there is no provably correct thing to do
  - Sometimes you must react quicker without time for reasoning
- Rational behavior: doing the right thing
- The right thing:
  - that which is expected to maximize goal achievement,
  - given the available information
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action

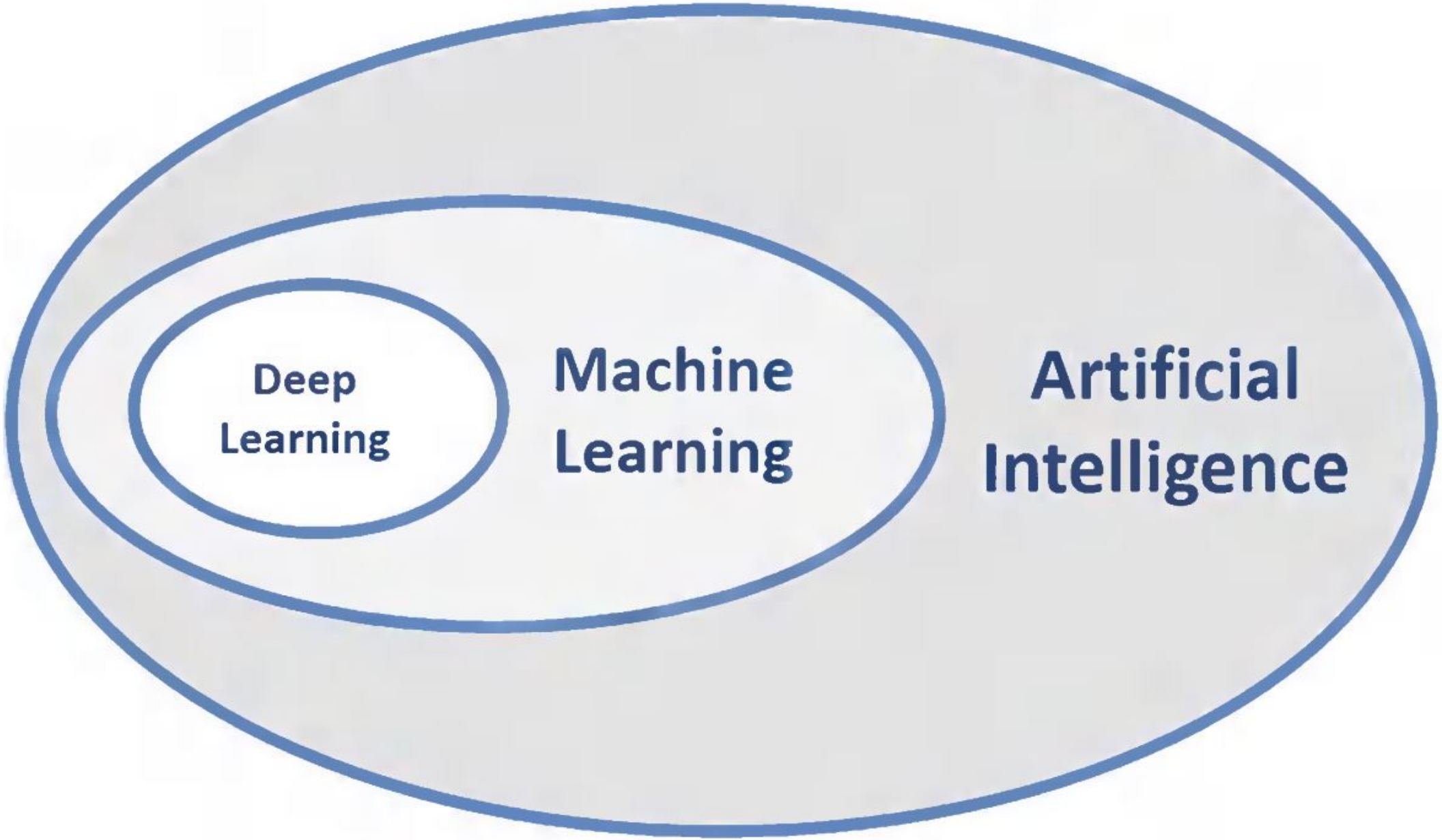

sensors
percepts
environment
?
agent
actions
actuators


How Does
A RATIONAL AGENT WORK?
01
Perception
02
Interpretation and Knowledge Update
03
Decision-Making
04
Action
05
Feedback Loop

What is AI?

- Machines that replicate human-like thinking.
- Analyze vast data to make decisions.




Deep Learning
Machine Learning
Artificial Intelligence

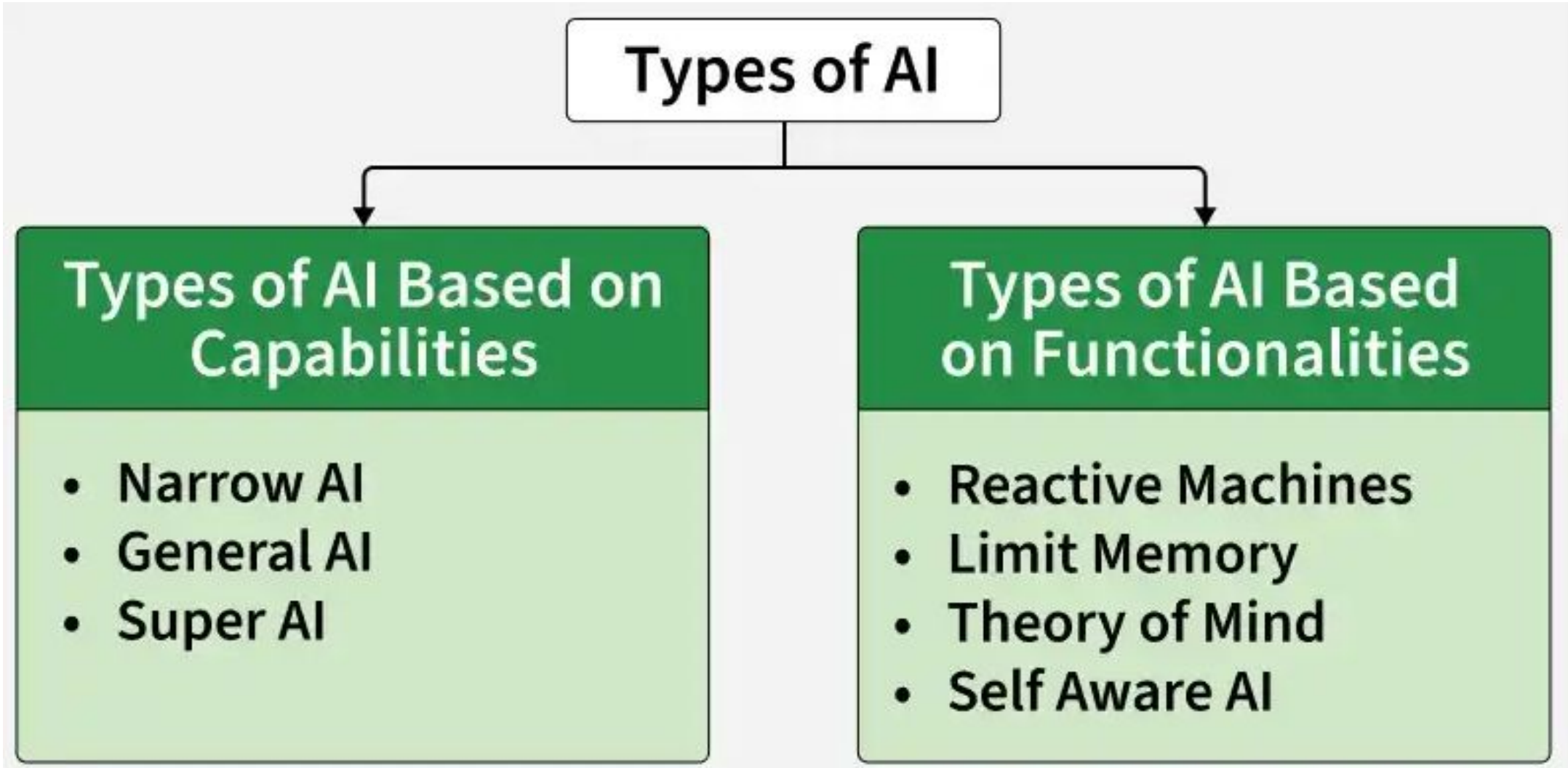
Artificial intelligence (AI)

- **Artificial intelligence (AI)** is a branch of computer science that develops computational systems capable of performing tasks typically requiring human intelligence. This includes **learning**, **reasoning**, **pattern recognition**, and **decision-making**.
- Modern AI powers everything from virtual assistants and web search engines to complex autonomous vehicles.

Artificial intelligence (AI)

- Artificial intelligence (AI) is a term used to describe the ability of computer systems to perform tasks that typically require human intelligence. AI systems use math and logic to simulate human reasoning, and can learn from their mistakes to improve their accuracy.
- The OECD AI Principles now define AI as: "**A machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.**"

(*OECD-Organization for Economic Co-operation and Development)


Types of AI
Types of AI Based on Capabilities
Narrow AI
General AI
Super AI
Types of AI Based on Functionalities
Reactive Machines
Limit Memory
Theory of Mind
Self Aware AI

Types of AI Based on Capabilities

1. Narrow AI (Weak AI)

Narrow AI is designed and trained on a specific task or a narrow range tasks. They performs their designated tasks but mainly lack in the ability to generalize tasks.

Examples:

1. Voice assistants like Siri, Alexa
2. Facial recognition in security
3. Recommendation engines like Netflix
4. Despite being highly efficient at specific tasks, they lacks the ability to function beyond its predefined scope. These systems do not possess understanding or awareness.

2. General AI (Strong AI)

General AI refers to machines that can perform any intellectual task like humans, with the ability to learn and adapt across tasks, though it remains theoretical and still not fully developed.

Potential Applications:

1. Robots learning new skills autonomously
2. AI diagnosing complex medical issues
3. Systems capable of coding, cooking or driving

3. Super AI (Super Intelligent AI)

Super AI is a theoretical concept where AI surpasses human intelligence. They are able to make decisions of their own and solve problem of its own.

Potential Applications:

1. Outperforms humans in all fields
2. Can be creative and make decisions
3. Raises ethical and control concerns

Types of AI Based on Functionalities
1. Reactive Machines

Reactive machines purely operates based on the present data and do not store any previous experiences or learn from past actions. These systems respond to specific inputs with fixed outputs and are unable to adapt.

Examples:

1. Deep Blue makes Chess moves based on board patterns without learning from previous games.
2. AlphaGo is a Go-laying AI, predicts moves using pattern recognition without retaining memory from past games.

2. Limited Memory in AI

Limited Memory AI uses past data to make better decisions and predictions but lacks long-term memory, and most modern AI applications belong to this type.

Examples:

1. Self-driving cars observe traffic and road conditions to make safe driving decisions.
2. Chatbots can remember recent conversations to improve the flow and relevance of replies.

3. Theory of Mind

Theory of Mind AI seeks to understand human emotions, beliefs, and intentions, enabling more sophisticated and responsive interactions.

Potential Applications:

1. Human-robot interaction detecting emotions
2. Collaborative robots in healthcare adapting to patient needs

4. Self-Awareness AI

Self-Aware AI is an advanced AI that possesses consciousness, enabling it to understand emotions and have self-awareness like humans.

Potential Applications:

1. Fully autonomous moral decision-making systems
2. AI pursuing goals independently based on understanding of the environment

Core Concepts(Modern Real-World AI Systems)

1. Machine Learning (ML)

It is a subset of AI that enables systems to learn from data and improve performance without explicit programming.(Email Spam Filters)

2. Generative AI

It focuses on creating new content such as text, images, audio, and videos using learned patterns from large datasets.(AI image generators)

3. Natural Language Processing (NLP)

It enables computers to understand, interpret, and generate human language naturally.(Language translation)

4. Expert Systems

It simulate human decision-making using predefined rules and domain-specific knowledge.

5. Agentic AI

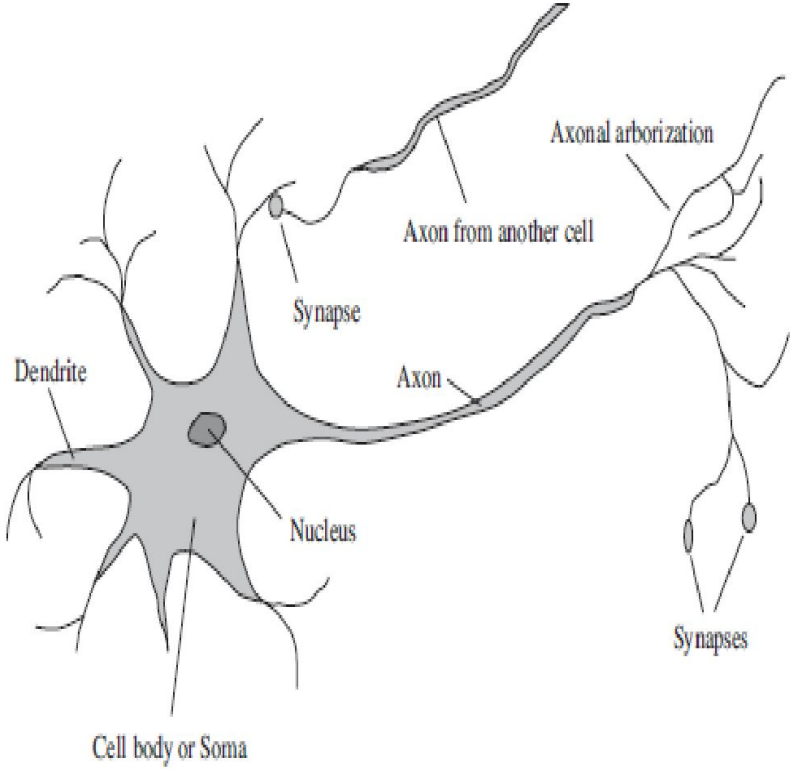
It acts autonomously to achieve goals making decisions and executing tasks without constant human input. It can plan, execute and adapt.(AI that books tickets after comparing prices)

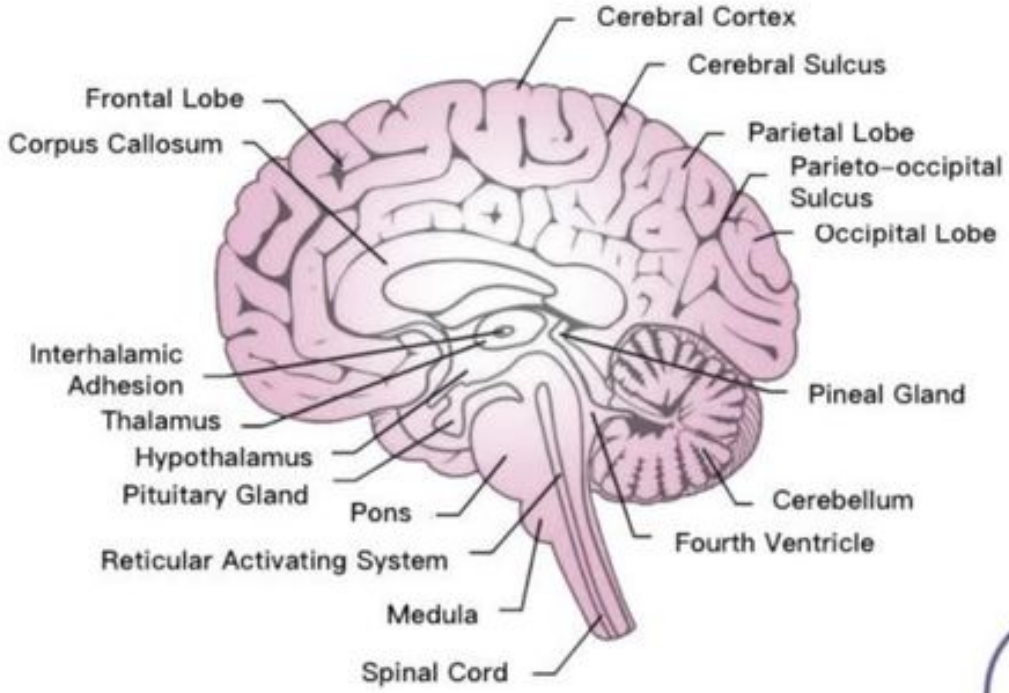
6. Computer Vision

It enables machines to analyze, recognize and interpret images and videos. Detects objects, faces and patterns from visuals.(Face recognition)

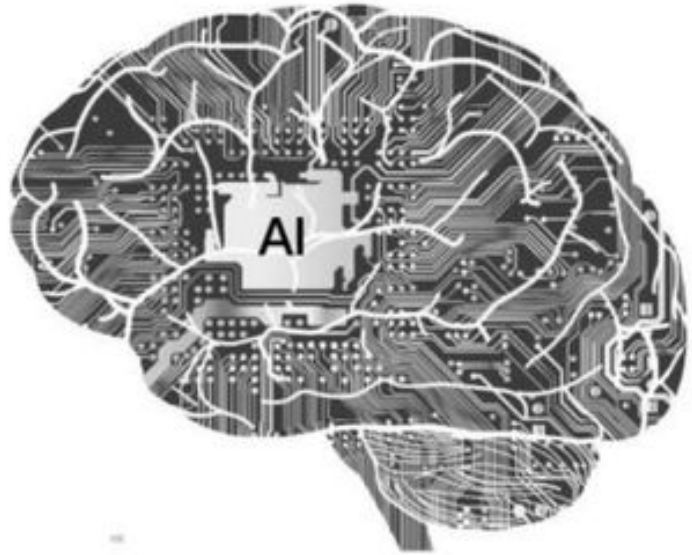
Foundations of Artificial Intelligence

- Neural science (physical substrate for mental activity):
  - How do brains process information?
    1. Parts of a nerve cell or neuron.
    2. Each neuron consists of a cell body, or soma, that contains a cell nucleus.
    3. Branching out from the cell body are a number of fibers called dendrites
    4. a single long fiber called the axon which stretches out for a long distance
    5. Typically, an axon is 1 cm long (100 times the diameter of the cell body), but can reach up to 1 meter.
    6. A neuron makes connections with 10 to 100,000 other neurons at junctions called synapses.
    7. Signals are propagated from neuron to neuron by a complicated electrochemical reaction.
    8. The signals control brain activity in the short term and also enable long-term changes in the connectivity of neurons.
    9. These mechanisms are thought to form the basis for learning in the brain.
    10. Most information processing goes on in the cerebral cortex, the outer layer of the brain.
    11. The basic organizational unit appears to be a column of tissue about 0.5 mm in diameter, containing about 20,000 neurons and extending the full depth of the cortex about 4 mm in humans).

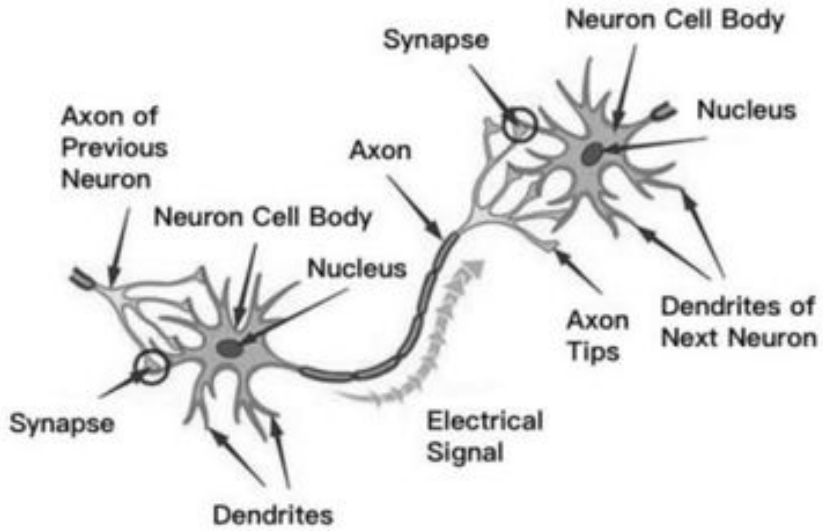


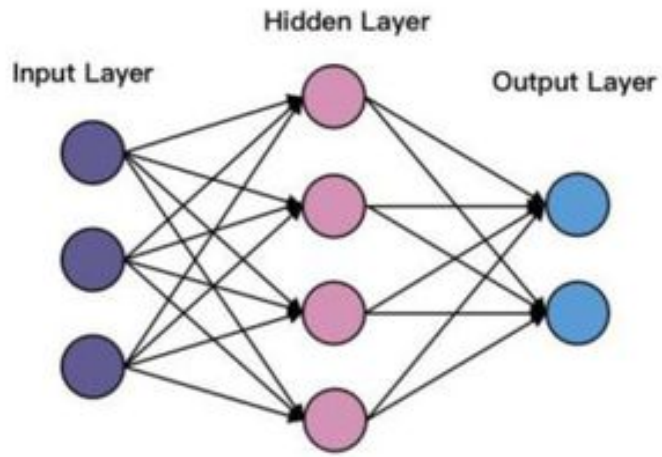


Human Intelligence



Artificial Intelligence



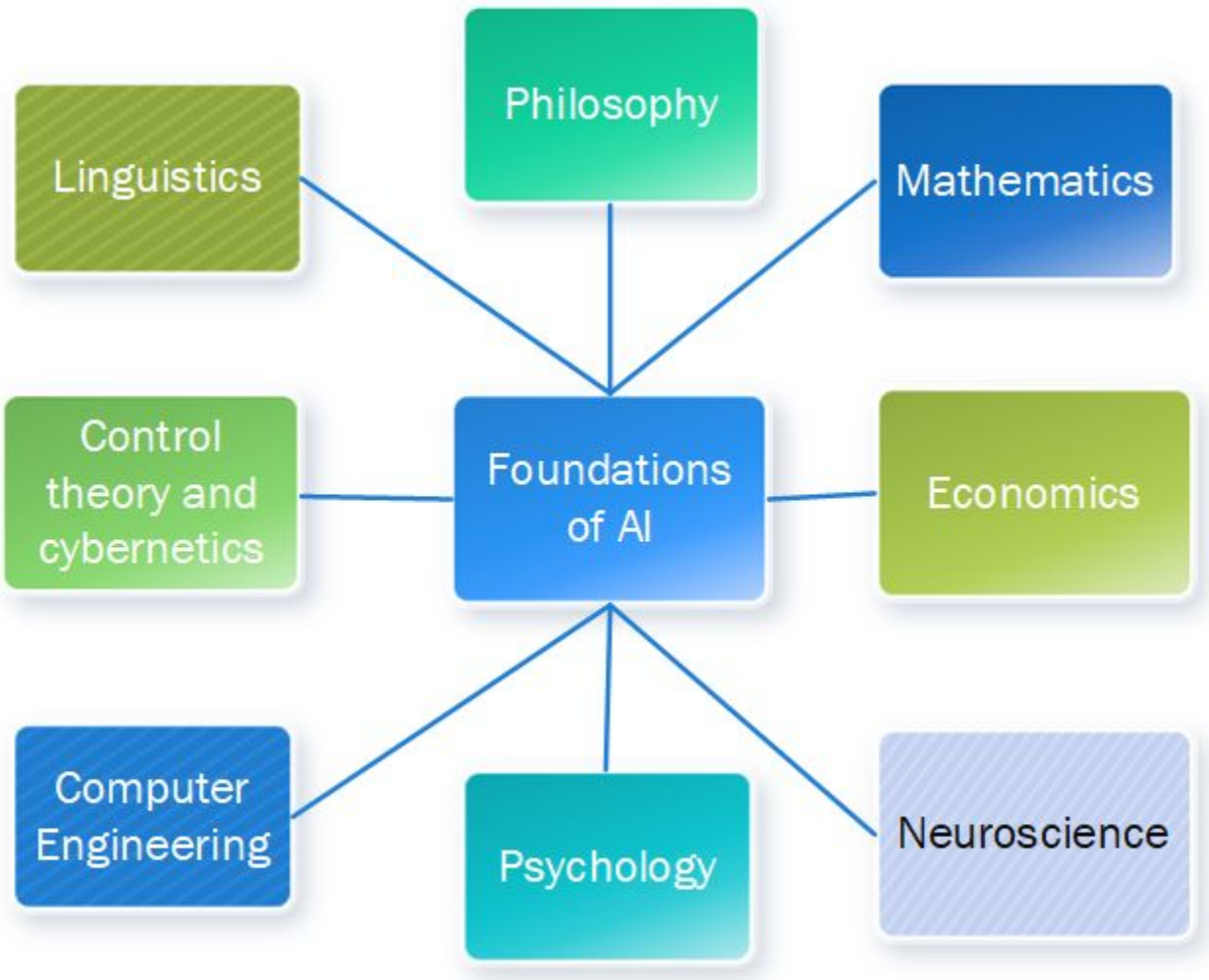


Human Intelligence vs Artificial Intelligence

- The diagram compares **Human Intelligence (HI)** with **Artificial Intelligence (AI)** and shows how AI is inspired by the human brain.
- **1. Human Intelligence**
  1. Human intelligence comes from the **human brain**, which consists of billions of neurons.
  2. The brain performs tasks such as **thinking, learning, reasoning, memory, creativity, and decision-making**.
  3. Neurons communicate through **electrical signals and synapses**, enabling humans to learn from experience.
- **2. Artificial Intelligence**
  1. Artificial Intelligence is the **simulation of human intelligence by computers**.
  2. AI uses **Artificial Neural Networks (ANNs)**, which are inspired by biological neurons.
  3. An ANN consists of an **Input Layer**, one or more **Hidden Layers**, and an **Output Layer** to process data and make predictions.

Human Intelligence	**Artificial Intelligence**
Natural intelligence	Machine-created intelligence
Learns from experience	Learns from data
Has emotions and creativity	No emotions or consciousness
Slower calculations	Very fast computations
Can solve diverse problems	Best for specific tasks

Foundations of Artificial Intelligence



Foundations of Artificial Intelligence

- **AI has roots in a number of scientific disciplines**
  - **Philosophy** (Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality):
    - Can formal rules be used to draw valid conclusions?
    - How does the mind arise from a physical brain?
    - Where does knowledge come from?
    - How does knowledge lead to action?
  - **Mathematics** (logic, algorithms, optimization, Formal representation, computation, (un)decidability, (in)tractability, probability):
    - What are the formal rules to draw valid conclusions?
    - What can be computed?
    - How do we reason with uncertain information?

- **Economics** (utility, decision theory):
  - How should we make decisions so as to maximize payoff?
  - How should we do this when others may not go along?
  - How should we do this when the payoff may be far in the future?
- **Neuroscience and Cognitive Science**
  - Researchers have learned how to reproduce specific cognitive processes in AI systems by examining the human brain. By providing a blueprint of human learning, memory, and decision-making, cognitive science and neuroscience help build more "human-like" robots, which form the basis of artificial intelligence.

- **Psychology**
  - Explains **human learning, memory, perception, cognition, and behavior**.
  - AI uses these concepts to improve learning algorithms and human-computer interaction.
  - Example: Intelligent tutoring systems and conversational agents.
- **Computer Engineering**
  - Provides the **hardware and software infrastructure** needed to build AI systems.
  - Includes programming languages, processors, GPUs, cloud computing, and databases.
  - Enables efficient implementation of AI algorithms.

- **7. Control Theory and Cybernetics**
  - Focuses on **feedback, automation, and system control**.
  - Helps AI systems monitor their environment and adjust their actions automatically.
  - Example: Robotics, autonomous vehicles, and industrial automation.
- **8. Linguistics**
  - Studies **human language structure, grammar, syntax, and semantics**.
  - Forms the basis of **Natural Language Processing (NLP)**, enabling AI to understand and generate human language.
  - Example: Chatbots, machine translation, and voice assistants.

MILESTONES

History of Artificial Intelligence

1950

Turing Test

Alan Turing proposes a test for machine intelligence.

1956

Dartmouth Conference

The term “Artificial Intelligence” is coined.

1966

ELIZA

First natural-language conversation program.

1980

Expert Systems

Rule-based systems used in industry & medicine.

1997

Deep Blue

IBM’s chess engine defeats world champion Kasparov.

2011

IBM Watson

Watson wins Jeopardy! against human champions.

2012

Deep Learning Revolution

CNNs achieve breakthrough image-recognition results.

2022

ChatGPT

Generative AI reaches mainstream, global audiences.

2024–26

Generative & Agentic AI

Multimodal, agentic AI systems automate real tasks.

History of Artificial Intelligence

- **1. The birth of AI (1943 – 1956)**
  - Pitts and McCulloch (1943):
    - simplified mathematical model of neurons (resting/firing states) can realize all propositional logic primitives (can compute all Turing computable functions)
  - Allen Turing:
    - Turing machine and Turing test (1950)
  - Claude Shannon:
    - information theory; possibility of chess playing computers

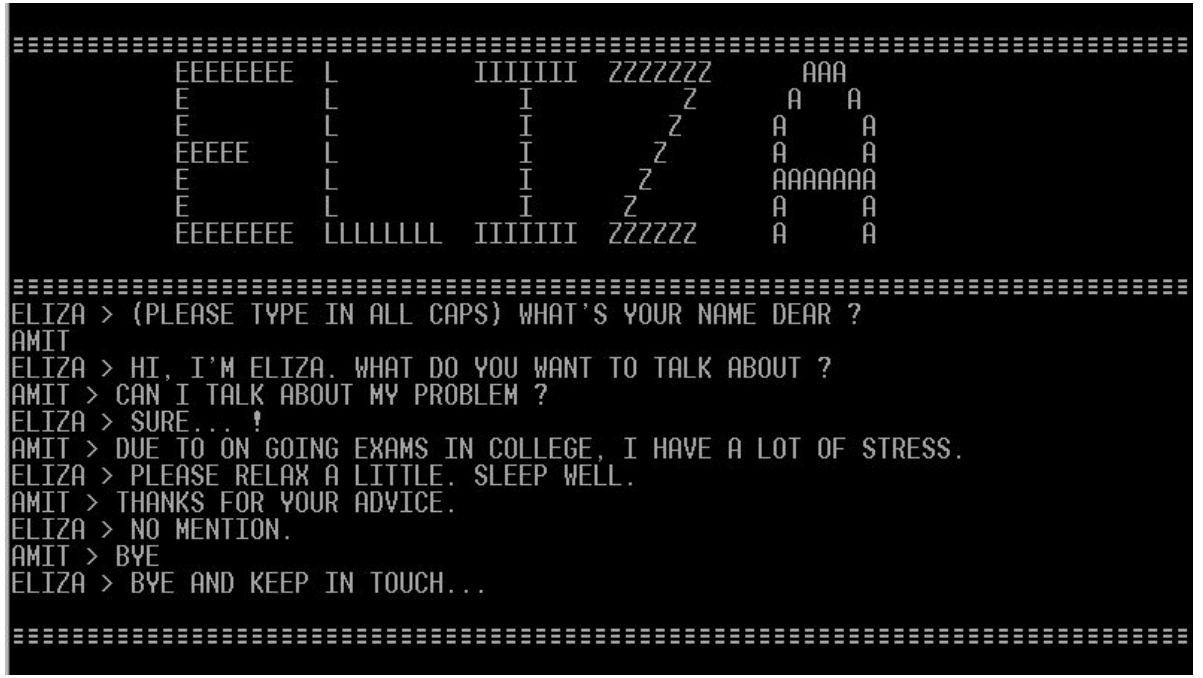
2. 1956 – Dartmouth Conference

What happened?

- The **Dartmouth Summer Research Project on Artificial Intelligence** was organized by:
  - John McCarthy
  - Marvin Minsky
  - Claude Shannon
  - Nathan Rochester
- This conference officially introduced the term
- **Artificial Intelligence (AI).**
- **Objective**
  - Researchers believed that every aspect of intelligence could eventually be described precisely enough for a machine to simulate it.
- **Importance**
  - Birth of Artificial Intelligence as an academic discipline.
  - Universities started AI laboratories.
  - AI research funding increased.

3. 1966 – ELIZA

- **Developed by**
  - Joseph Weizenbaum (MIT)
- **What is ELIZA?**
  - ELIZA was the world's first chatbot.
  - It simulated conversations using simple pattern matching.
- Example
  - **User:** I am feeling sad.
- **ELIZA:** Why are you feeling sad?
  - Although it did not understand language, people felt they were talking to a real therapist.
- **Importance**
  - Beginning of Natural Language Processing (NLP).
  - Foundation for modern chatbots.
  - Demonstrated human-computer interaction.



1980 – Expert Systems

- **What are Expert Systems?**
  - Expert Systems imitate the decision-making ability of human experts.
  - They use
  - IF-THEN rules
  - Knowledge Base
  - Inference Engine
  - Example
  - IF Fever = Yes AND Cough = Yes THEN Disease = Flu
- **Applications**
  - Medical diagnosis
  - Banking
  - Engineering
  - Manufacturing
- **Importance**
  - Expert systems became the first commercially successful AI applications.



1997 – Deep Blue

- **Developed by**
  - IBM
- **Achievement**
  - IBM's Deep Blue defeated the world chess champion
- **Garry Kasparov**
  - This was the first time a computer defeated a reigning world champion under standard tournament conditions.
- **Technology Used**
  - Massive search algorithms
  - Game tree pruning
  - Evaluation functions
  - Parallel computing
- **Importance**
  - Demonstrated machine superiority in structured games.
  - Boosted confidence in AI research.



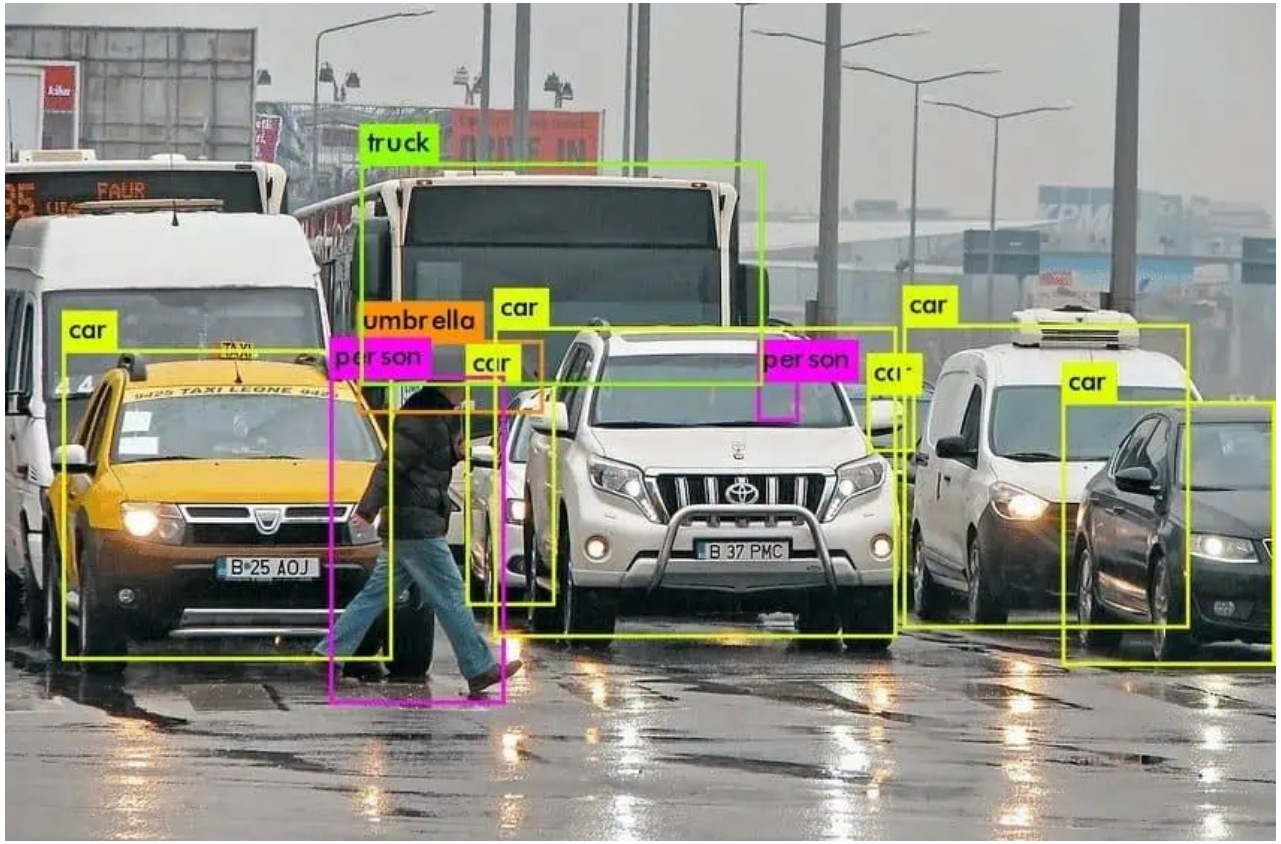
2011 – IBM Watson

- **What is Watson?**
  - IBM Watson is an AI question-answering system.
- **Achievement**
  - Watson defeated the two greatest Jeopardy! champions
  - Ken Jennings
  - Brad Rutter
- **Technologies**
  - Natural Language Processing
  - Information Retrieval
  - Machine Learning
- **Applications Today**
  - Healthcare
  - Business analytics
  - Customer support
- **Importance**
  - Showed AI could understand natural language and answer complex questions.



2012 – Deep Learning Revolution

- **What happened?**
  - Deep Learning became highly successful after the **AlexNet** neural network dramatically improved image recognition performance in the ImageNet competition.
- **Technology**
  - Artificial Neural Networks
  - Convolutional Neural Networks (CNN)
  - GPUs
  - Big Data
- **Achievements**
  - Image recognition
  - Face detection
  - Speech recognition
  - Self-driving cars
- **Importance**
  - Started the modern AI revolution.
  - Almost every AI system today uses deep learning.

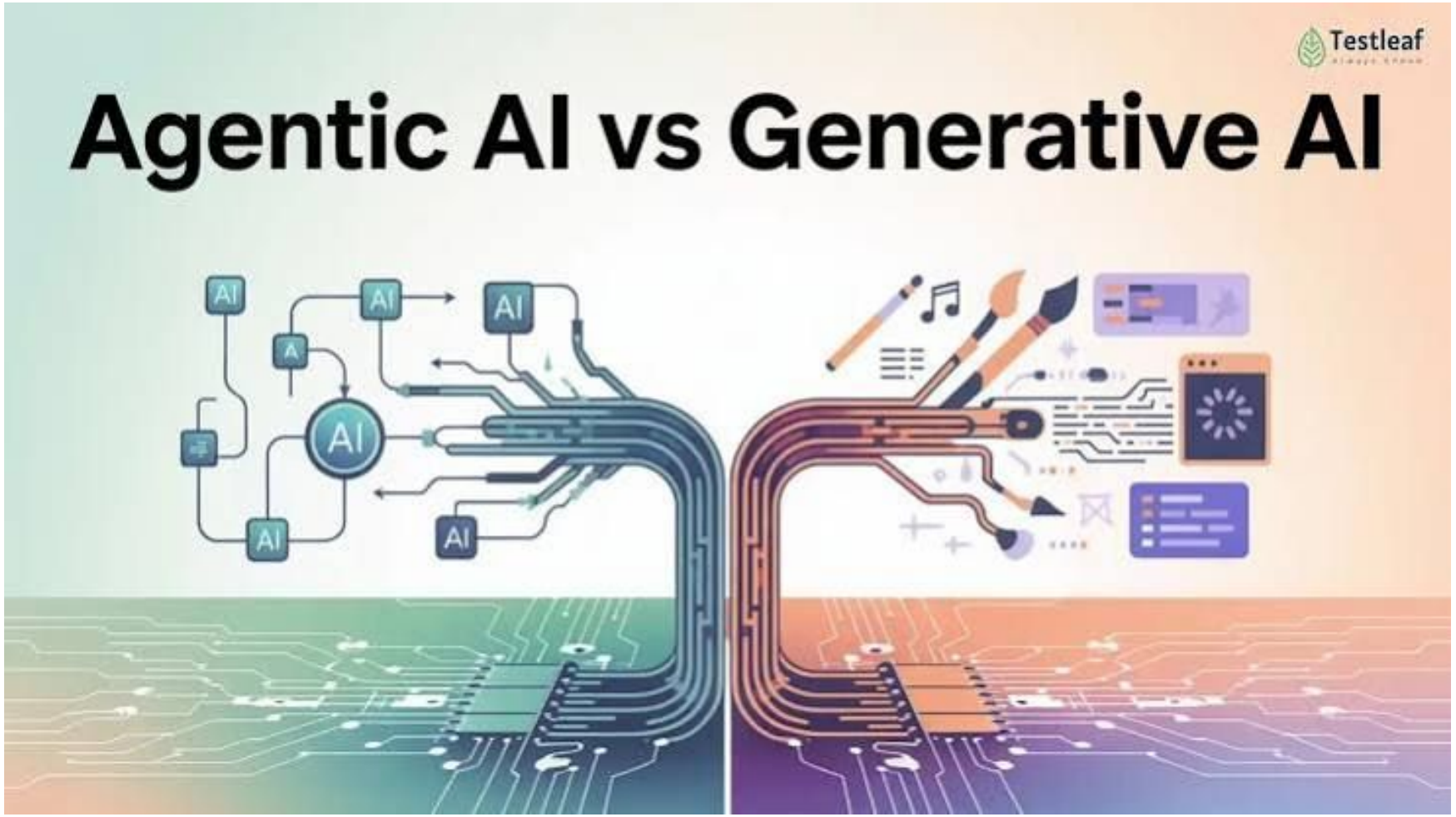


2022 – ChatGPT

- **Developed by**
  - OpenAI
- **What is ChatGPT?**
  - ChatGPT is a Generative AI model based on Large Language Models (LLMs).
  - It can
    - Answer questions
    - Write code
    - Generate essays
    - Summarize documents
    - Translate languages
    - Assist in education
- **Technology**
  - Transformer Architecture
  - GPT (Generative Pre-trained Transformer)
  - Reinforcement Learning from Human Feedback (RLHF)
- **Importance**
  - Made Generative AI accessible to millions of users and transformed education, business, software development, and content creation.



2024–2026 – Generative & Agentic AI



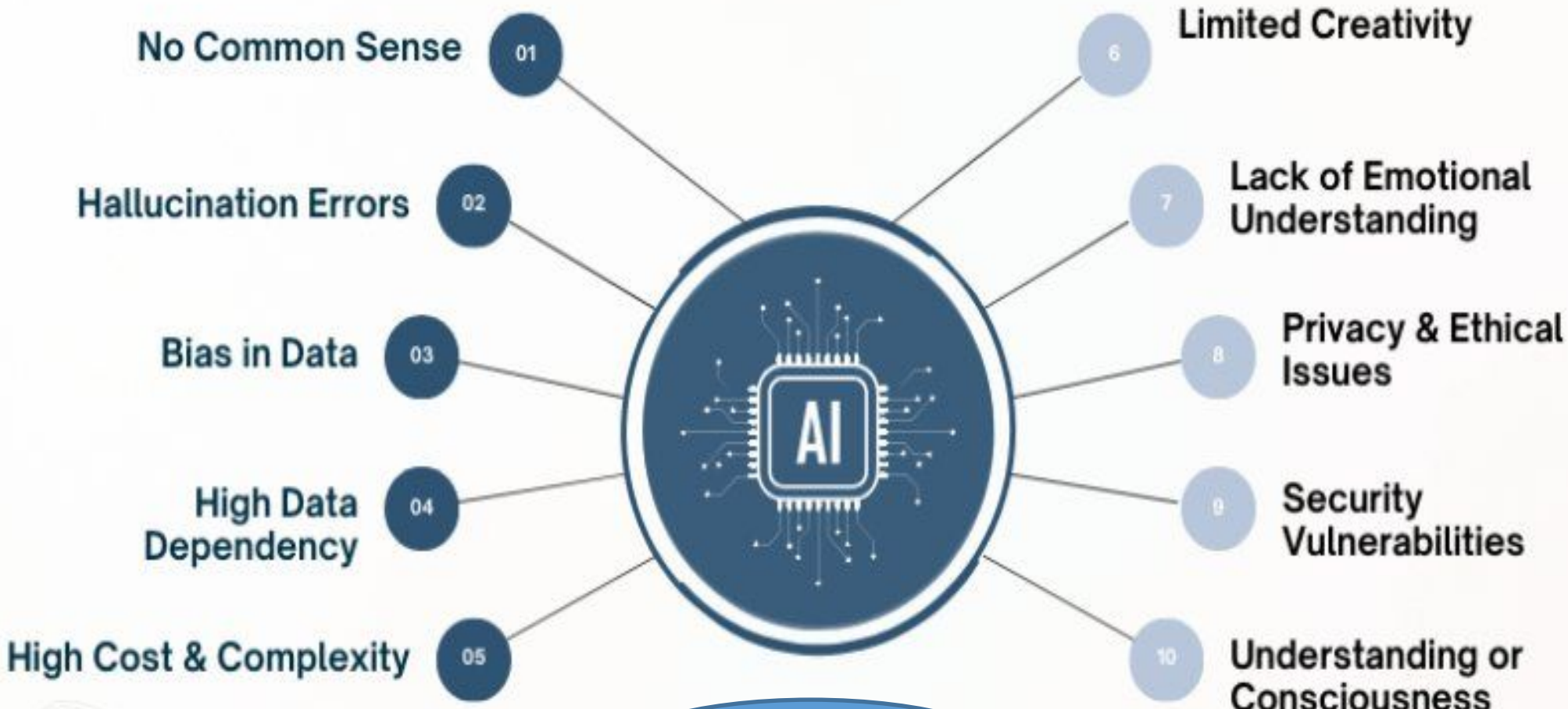
Generative AI

- Generative AI creates new content such as:
  - Text
  - Images
  - Audio
  - Video
  - Code
- Examples
  - ChatGPT
  - DALL·E
  - Gemini
  - Claude
  - Midjourney

Agentic AI

- Agentic AI goes beyond generating content. It can:
  - Understand goals
  - Plan tasks
  - Use tools
  - Make decisions
  - Execute multi-step workflows
  - Learn from feedback
- Examples
  - AI research assistants
  - Autonomous coding agents
  - AI customer service agents
  - AI workflow automation

Top 10 Limitations of Artificial Intelligence (2026)



Limits of AI

1. AI Lacks Common Sense

- AI can beat world chess champions, but it cannot answer simple reasoning questions reliably.
- **Example:**
  Ask an AI "If I put a glass on the edge of a table and push it slightly, what happens?"
  - It may answer correctly…
    or it may confidently hallucinate something entirely different.
  - Humans learn common sense through lived experience.
    AI learns patterns through data.
- **Why this is a problem:**
  - AI struggles with unexpected situations
  - Self-driving cars misinterpret rare events
  - Chatbots fail at basic reasoning
  - Robots cannot handle unpredictable tasks
- **Future Fix:**
  - Researchers are working on **Common Sense AI frameworks**, but we are still far from giving machines real-world intuition.

2. AI Hallucinations (False Information)

- **Hallucination** occurs when AI generates **confident but incorrect or imaginary information**.
- **Examples**
  - Fake laws or legal sections
  - Fake statistics or data
  - Fake people or biographies
  - Incorrect medical facts
  - Fake quotes
  - Non-existent URLs or references
- **Why Does It Happen?**
  - AI predicts the **most likely text based on patterns**, not verified facts.
- **Risks**
  - Wrong medical advice
  - Incorrect legal information
  - Misleading financial data
  - Fake news
  - Errors in education and research

3. Bias in AI Systems

- AI can learn and repeat **biases present in its training data**, leading to unfair decisions.
- **Examples:**
  - Hiring systems favoring one gender
  - Face recognition performing poorly on darker skin tones
  - Biased credit scoring
  - Inaccurate healthcare predictions for minority groups
  - **Future Fix:** Use diverse datasets, fairness algorithms, bias audits, and ethical AI frameworks.

4. AI is Extremely Data-Dependent

- AI requires **large amounts of high-quality data** to learn and make accurate predictions. Poor or biased data leads to poor results.
- **Problems:**
  - Incorrect predictions
  - Bias
  - Hallucinations
  - Unreliable outputs
- **Future Fix:** Self-learning models, synthetic data, human-guided training, and better data labeling.

5. High Cost of Development and Maintenance

- Developing AI requires **powerful hardware, cloud computing, and continuous updates**, making it expensive.

- **Problems:**
  - High development cost
  - Small companies cannot afford advanced AI
  - Limited access to AI technology
- **Future Fix:** Smaller AI models, model distillation, open-source AI, and efficient AI chips.

6. Limited Creativity

- AI generates content by **learning patterns from existing data** rather than creating from imagination or emotions.
- **Examples:**
  - AI-generated art
  - AI-composed music
  - AI-written content
- **Future Fix:** Better generative models, though true human creativity cannot be replicated.

7. Lack of Emotional Intelligence

- AI can simulate empathy but **cannot truly understand or feel emotions**.
- **Examples:**
  - AI counselors cannot replace therapists
  - Robotic customer support
  - Emotionally shallow responses
- **Future Fix:** Emotional AI may improve, but genuine human emotions cannot be achieved.

8. Privacy Risks

- AI often collects **personal data**, creating privacy concerns.
- **Examples:**
  - Voice assistants recording conversations
  - Data leaks
  - Facial recognition surveillance
  - Identity theft
- **Future Fix:** Privacy-first AI, local processing, strong encryption, and stricter regulations.

9. Security Vulnerabilities

- AI systems can be **attacked or manipulated** by hackers.
- **Examples:**
  - Adversarial attacks
  - Deepfakes
  - Model poisoning
  - Prompt injection attacks
- **Future Fix:** Secure AI models, adversarial training, and authentication for AI-generated content.

10. No Explainability (Black Box Problem)

- Many AI models make decisions **without explaining how they reached them**.
- **Problems:**
  - Lack of trust
  - Legal and ethical issues
  - Difficult to verify decisions
- **Future Fix:** Explainable AI (XAI) to make AI decisions more transparent.

11. Job Displacement Concerns

- AI automates **repetitive and routine tasks**, affecting some jobs.
- **Examples:**
  - Administrative work
  - Data entry
  - Basic customer support
- **Future Fix:** Focus on **AI + Human collaboration**, reskilling, and creating new job opportunities.

12. AI Cannot Understand Context Like Humans

- AI struggles to understand **sarcasm, humor, emotions, cultural references, and real-life situations**.

- **Example:** It cannot fully understand the request, *"Can you get me something to eat?"* without additional context.
- **Future Fix:** Improved context-aware models, though human-level understanding remains challenging.

13. AI Lacks Real Consciousness or Understanding

- AI **does not think, feel, or possess self-awareness**. It only simulates intelligence by recognizing patterns.

- **Limitation:** AI has no consciousness, intentions, or genuine understanding.
- **Future Fix:** Currently, there is **no technology capable of giving AI true consciousness**.

Real-World Examples Where AI Failed

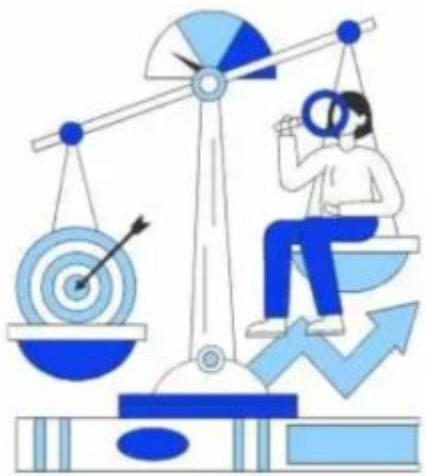
- **1. Self-driving Car Confusion**
  - AI misread a truck's white surface as sky → fatal accident.
- **2. Recruitment AI Bias**
  - AI hired more men because historical data favored male applicants.
- **3. Medical AI Failure**
  - AI misdiagnosed darker-skinned patients due to biased training data.
- **4. Image Recognition Mistakes**
  - AI labeled random people as criminals due to dataset flaws.
- These failures show why AI still needs human oversight.

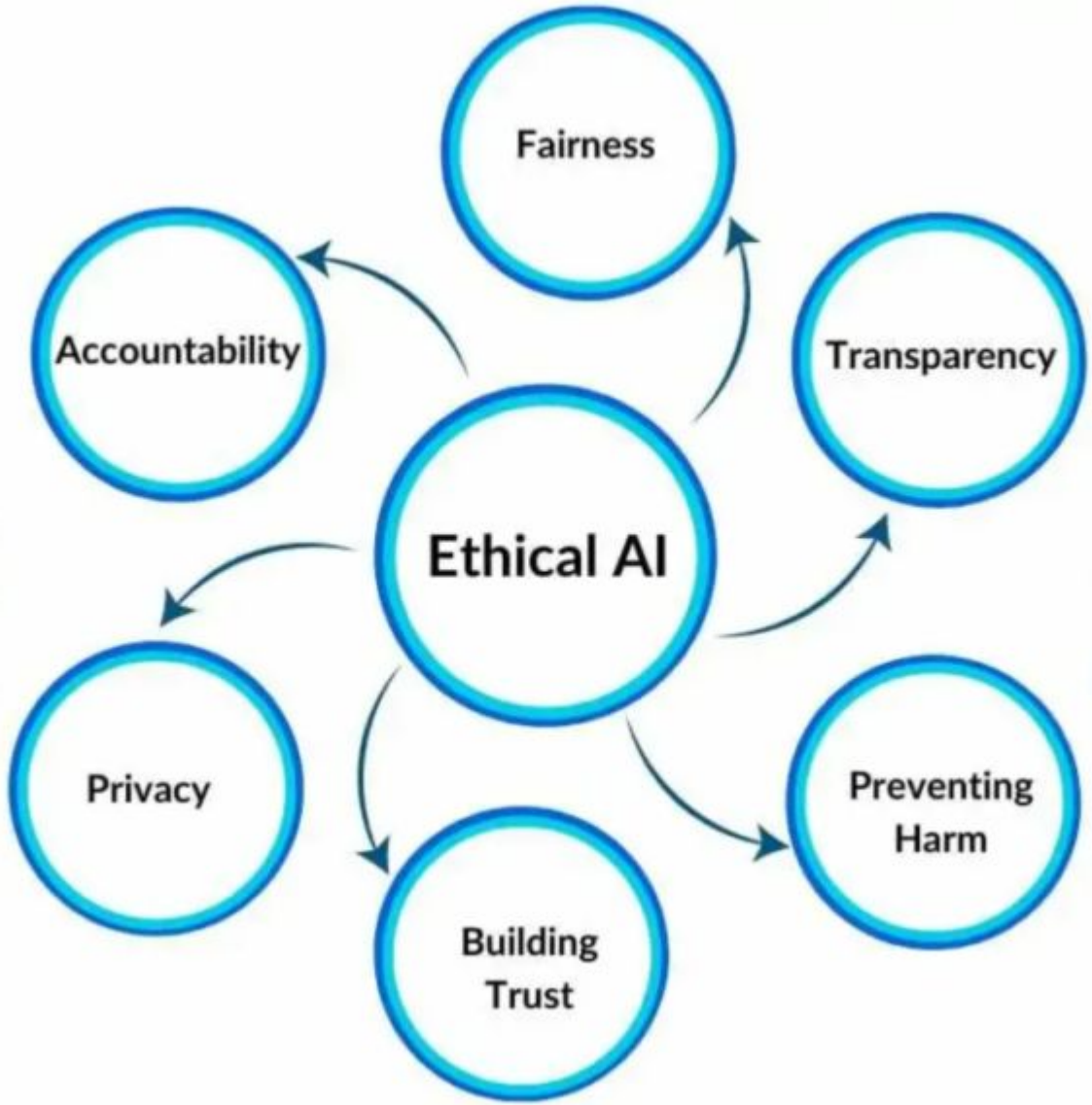
Future Fixes: How AI's Limitations Will Improve in the Next Decade

1. **Better Common Sense Models**
   Using simulations, robotics, and hybrid models.
2. **Hallucination-Free AI (Partially)**
   Real-time data + fact-checking + retrieval models.
3. **Ethical & Fair AI**
   More balanced datasets, audits, and fairness frameworks.
4. **Low-Cost AI Models**
   Efficient small models with near-big-model performance.
5. **Emotionally Aware AI**
   Better tone detection + sentiment models.
6. **Transparent AI (Explainability)**
   Visual reasoning maps that show how decisions were made.

But remember:

AI will always have limits because AI is not human.




Fairness
Accountability
Transparency
Ethical AI
Privacy
Preventing Harm
Building Trust



1. Fairness

- AI should treat everyone equally without discrimination.
- It should avoid bias based on gender, race, age, or religion.
- **Example:** A hiring AI should evaluate all candidates fairly.

2. Transparency

- AI decisions should be understandable and explainable.
- Users should know how and why an AI system makes decisions.
- **Example:** A bank should explain why an AI rejected a loan application.

3. Preventing Harm

- AI should not cause physical, financial, or social harm.
- Systems must be tested to reduce errors and misuse.
- **Example:** Self-driving cars should prioritize passenger and pedestrian safety.

4. Building Trust

- AI should be reliable, secure, and accurate so that users can trust its decisions.
- **Example:** Medical AI should provide consistent and dependable diagnoses.

5. Privacy

- AI should protect users' personal and sensitive data.
- It should collect only necessary information and keep it secure.
- **Example:** Healthcare AI should keep patient records confidential.

6. Accountability

- Organizations and developers must be responsible for AI decisions and any harm caused.
- AI should not replace human responsibility.
- **Example:** If an AI system makes a wrong decision, the organization deploying it is accountable

Future of AI

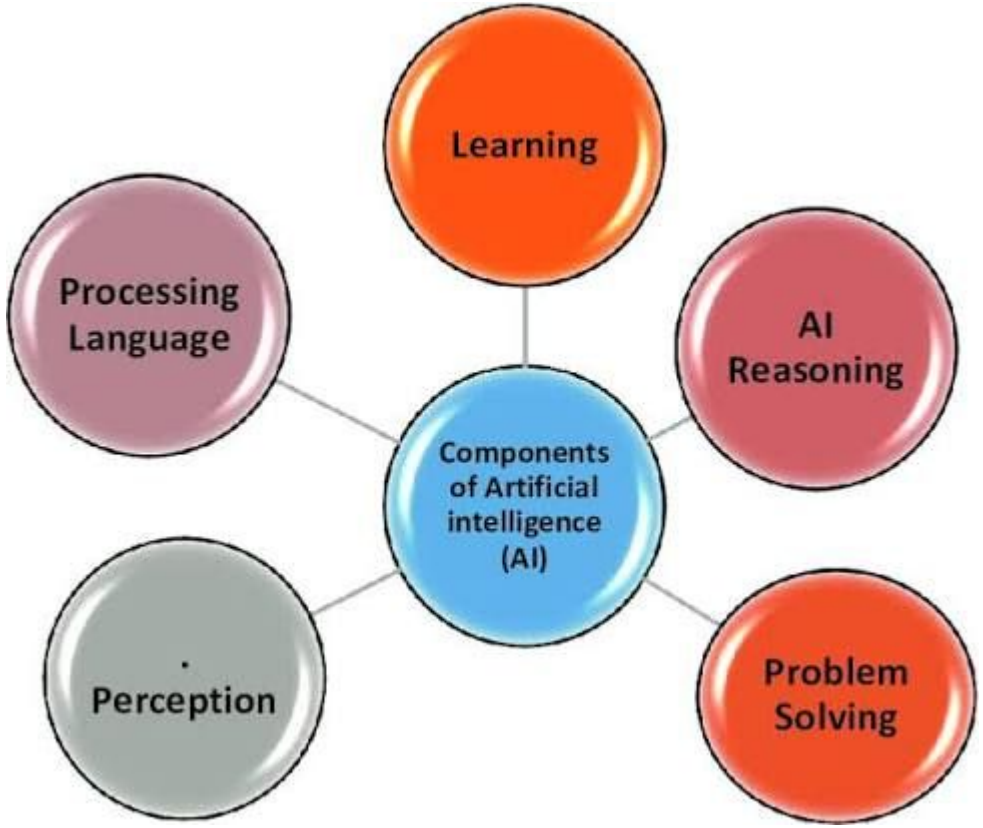
- The future of artificial intelligence (AI is a topic of great interest and speculation, and while it's challenging to predict with certainty, there are several trends and directions that are likely to shape the future of AI:
- **Continued Advancements in Machine Learning**:
  - Machine learning techniques, including deep learning, will continue to evolve, leading to AI systems that are more capable of learning from data and making sense of complex patterns.
- **AI in Healthcare**:
  - AI will play a significant role in healthcare, aiding in diagnosis, drug discovery, and personalized medicine. Telemedicine and remote health monitoring will become more common.
- **AI in Autonomous Systems**:
  - Autonomous vehicles, drones, and robots will become more prevalent, transforming transportation, logistics, and manufacturing.

- **Natural Language Processing (NLP)**:
  - NLP will continue to improve, making AI systems better at understanding and generating human language. This will impact customer service, content generation, and more.
- **AI in Education**:
  - AI will enhance personalized learning experiences, automate administrative tasks, and provide intelligent tutoring.
- **AI in Climate and Environmental Research**:
  - AI will be used to analyze vast amounts of environmental data, model climate change, and optimize resource management for sustainability.
- **AI Ethics and Regulation**:
  - The development of AI ethics and regulations will continue to address issues like bias, privacy, accountability, and the responsible use of AI.
- **Explainable AI (XAI)**:
  - Efforts to make AI more transparent and interpretable will grow, allowing users to understand how AI systems make decisions.

- **AI in Finance**:
  - AI will play a role in risk assessment, fraud detection, algorithmic trading, and personal finance management.
- **Quantum Computing and AI**:
  - The intersection of quantum computing and AI is promising, with the potential to solve complex problems that are currently intractable.
- **AI Assistants and Personalization**:
  - AI-powered virtual assistants will become more integrated into daily life, offering highly personalized recommendations and services.
- **AI for Creativity and Art**:
  - AI-generated art, music, and literature will become more sophisticated, blurring the line between human and machine creativity.
- **AI in Cybersecurity**:
  - AI will be used for threat detection, anomaly detection, and proactive defense against cyberattacks.

- **AI in Agriculture**:
  - AI will optimize farming practices, crop management, and precision agriculture, contributing to food security.
- **AI and Social Challenges**:
  - AI will play a role in addressing social issues, such as poverty, inequality, and public health, through data analysis and policy recommendations.
- **AI and Personal Wellness**:
  - AI-driven health and wellness applications, including mental health support, will continue to grow.
- **AI for Disaster Response**:
  - AI will aid in disaster prediction, response coordination, and recovery efforts.
- **AI in Entertainment and Gaming**:
  - AI will create more immersive and interactive gaming experiences and personalized content recommendations in the entertainment industry.
- **Global Collaboration and Ethical Development**:
  - Collaboration among countries and organizations will be essential to ensure that AI is developed and used ethically and for the benefit of humanity.

AI Components



AI Components

- **Learning**
- Learning in AI occurs when machines or computer systems memorize specific data or new material. Specifically, advancements in deep machine learning now enable enhancements in prescriptive and predictive analytics through the use of operational data.
- Machine learning can find hidden correlations in various data. With this information, the network can create a predictive model that is able to pinpoint future machine failures in manufacturing.
- Machine learning may even predict when the failure will occur. This can enable companies to know when and how many parts to order.

- **AI Reasoning**
- AI uses the ability to make inferences when applying reasoning based on commands it is given or other information at its disposal. For example, virtual assistants will offer restaurant recommendations based on the specific orders or questions it receives.
- The assistant will use reasoning to decide what restaurants to suggest based on the questions it received and the nearest location of various restaurants.
- This type of reasoning involves drawing inferences. Inferences include two categories: deductive and inductive reasoning.

- **Problem Solving**
- In the most basic of terms, an AI's problem-solving ability is based on the application and manipulation of data, where the solution needs to be x.
- Alternatively, in more advanced applications, problem-solving techniques in the context of AI can include the development of efficient algorithms and performing root cause analysis with the goal of discovering a desirable solution.
- AI implements heuristics when solving problems by devising a solution using trial and error techniques. Specific examples of problem-solving in AI would include the use of predictive technology in the area of online shopping.
- When a shopper is looking for a product and doesn't know the exact name of the product, AI can assist in dramatically reducing the possibilities.

- **Perception**
- Perception is when different sense organs, whether real or artificial, scan the environment. For example, AI scans the environment through sense components such as temperature sensors and cameras.
- Autonomous driving is an example of how AI implements perception. They are able to perceive and comprehend the environment around them, including traffic lights, road lines, and weather conditions.
- Other examples include a GPS system or smart speakers that respond to human queries. After capturing elements of the surrounding environment, a perceiver will analyze the different objects, extract their features, and analyze the relationships among them.

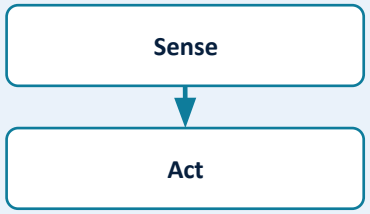
- **Processing Language**
- AI processes language in something as seemingly simple as spellcheck and autocorrect. Computer programs use neural networks to scan large bodies of text for misspelled words and language irregularities.
- Another way AI uses language processing is when it weeds out spam in email systems. For example, spam filters delegate specific messages as spam when seeing certain words or combinations of words.

AI Architecture

- Artificial Intelligence (AI) architecture refers to the structure and design of AI systems, which includes the components, layers, and interactions that enable the AI system to perform tasks, make decisions, and adapt to its environment. AI architectures can vary significantly based on the specific application and the AI approach being used.

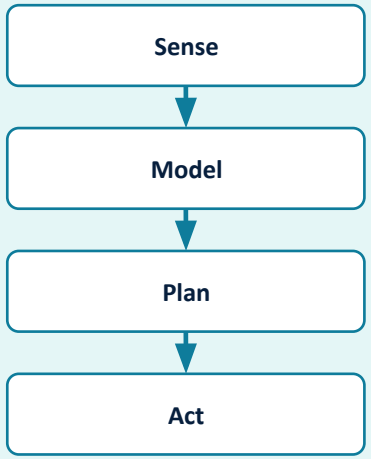
AI Architectures

Reactive



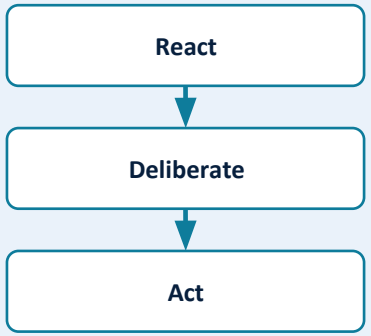
No memory of past states; responds only to current input.

Deliberative



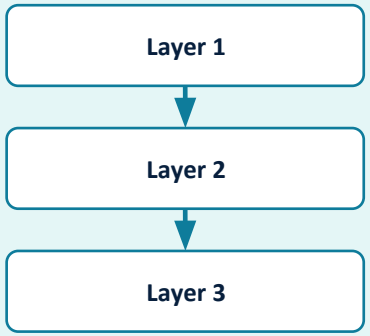
Builds an internal world model and reasons before acting.

Hybrid



Combines fast reflexes with slower deliberate planning.

Layered

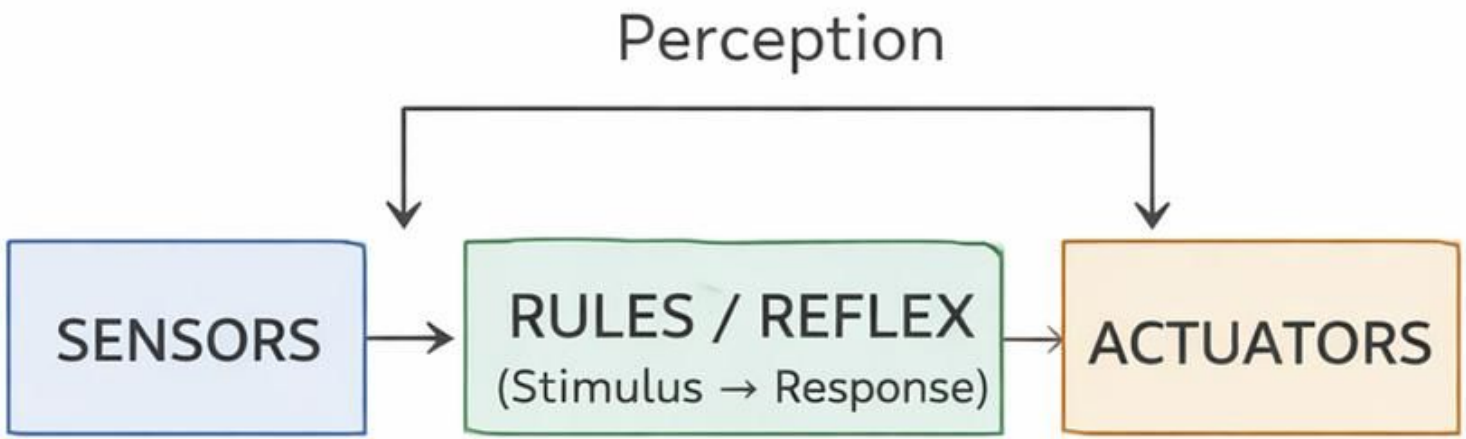


Each layer handles a different level of behaviour or control.

Multi-Agent



Several agents coordinate or compete to reach shared goals.



- Stateless
- No Memory
- No Planning

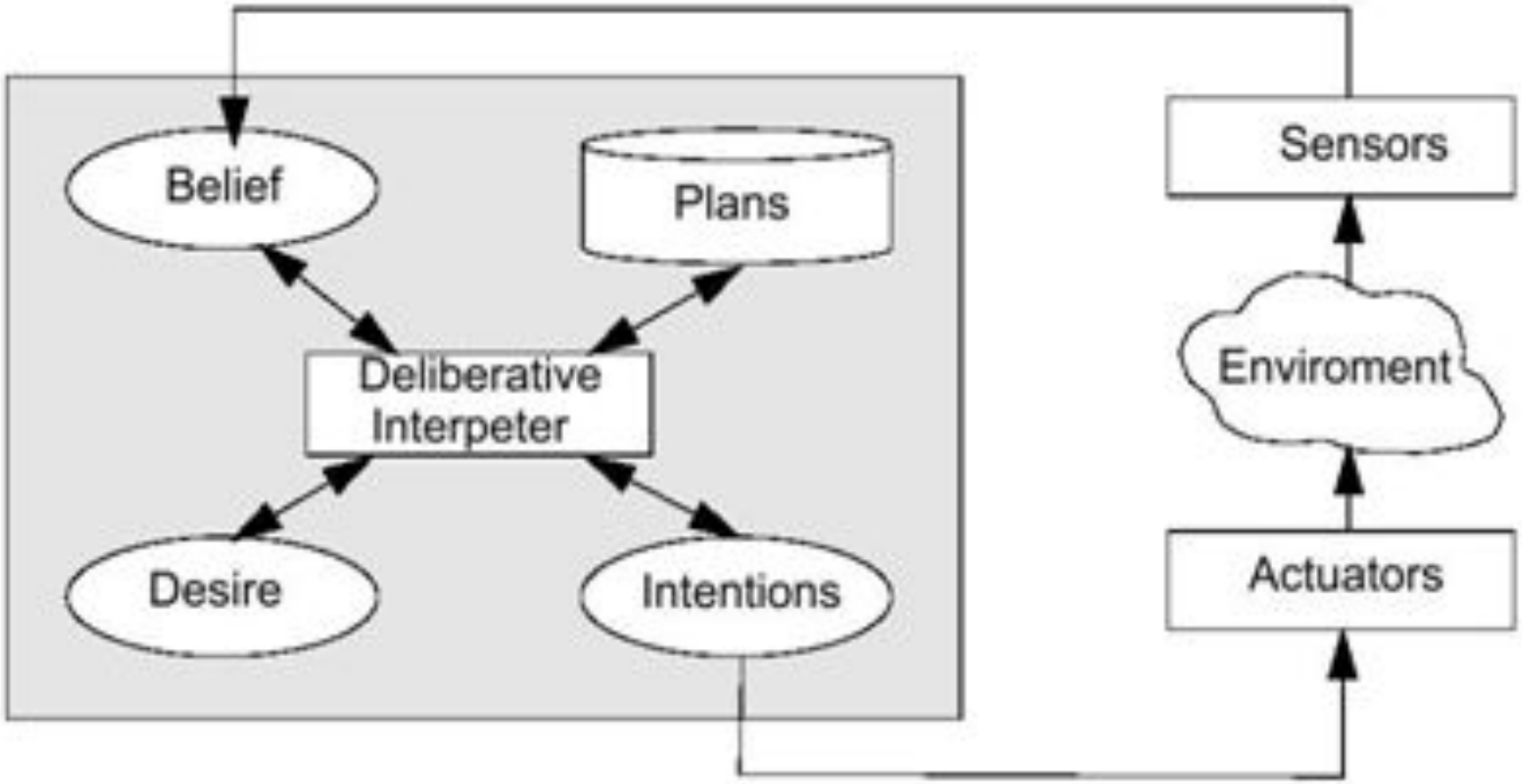
Reactive Agent

- Immediate stimulus → response
- No internal memory
- No planning or learning
- Deterministic behavior

1. Reactive architecture

- Reactive architecture is the simplest AI architecture. It responds directly to the current environment without maintaining memory or planning for the future.
- **Working**
- Receives input from sensors.
- Applies predefined rules.
- Produces an immediate action.
- No memory or internal world model.
- **Example**:- Automatic Sliding Door

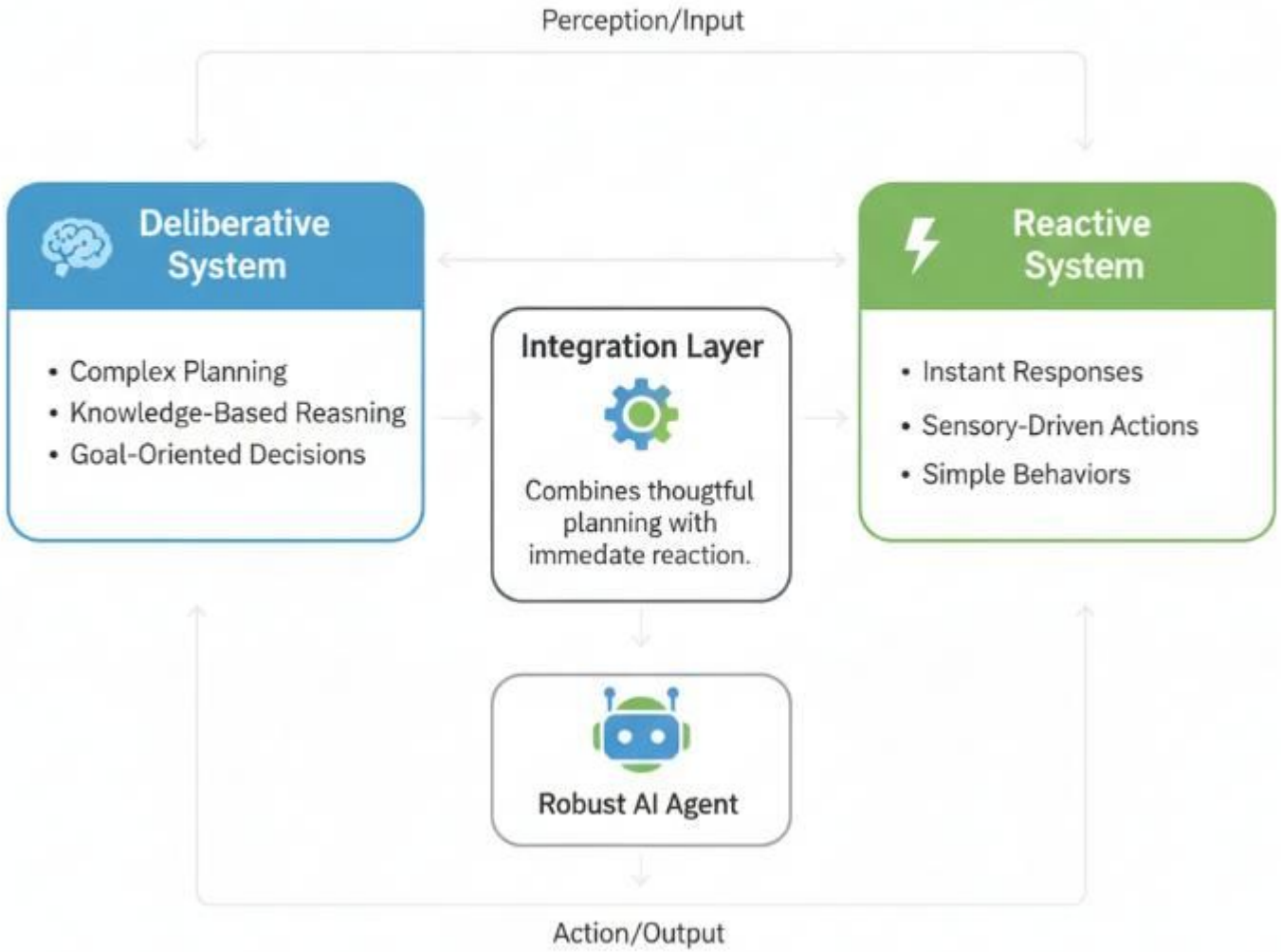
2. Deliberative Architecture



2. Deliberative Architecture

- **Definition**
- Deliberative architecture maintains an internal model (knowledge base) of the environment and plans before taking action.
- **Working**
- Collects environmental information.
- Updates internal knowledge.
- Plans the best action.
- Executes the selected action.
- **Example**:- GPS Navigation System

Hybrid AI Architecture



3. Hybrid Architecture

- **Definition**
- Hybrid architecture combines the speed of reactive systems with the intelligence of deliberative systems.
- **Working**
- Reactive layer handles urgent responses.
- Deliberative layer performs planning.
- Coordinator chooses appropriate actions.

Hybrid Architecture Examples

1. Delivery Robot

Avoids obstacles instantly (**Reactive**).

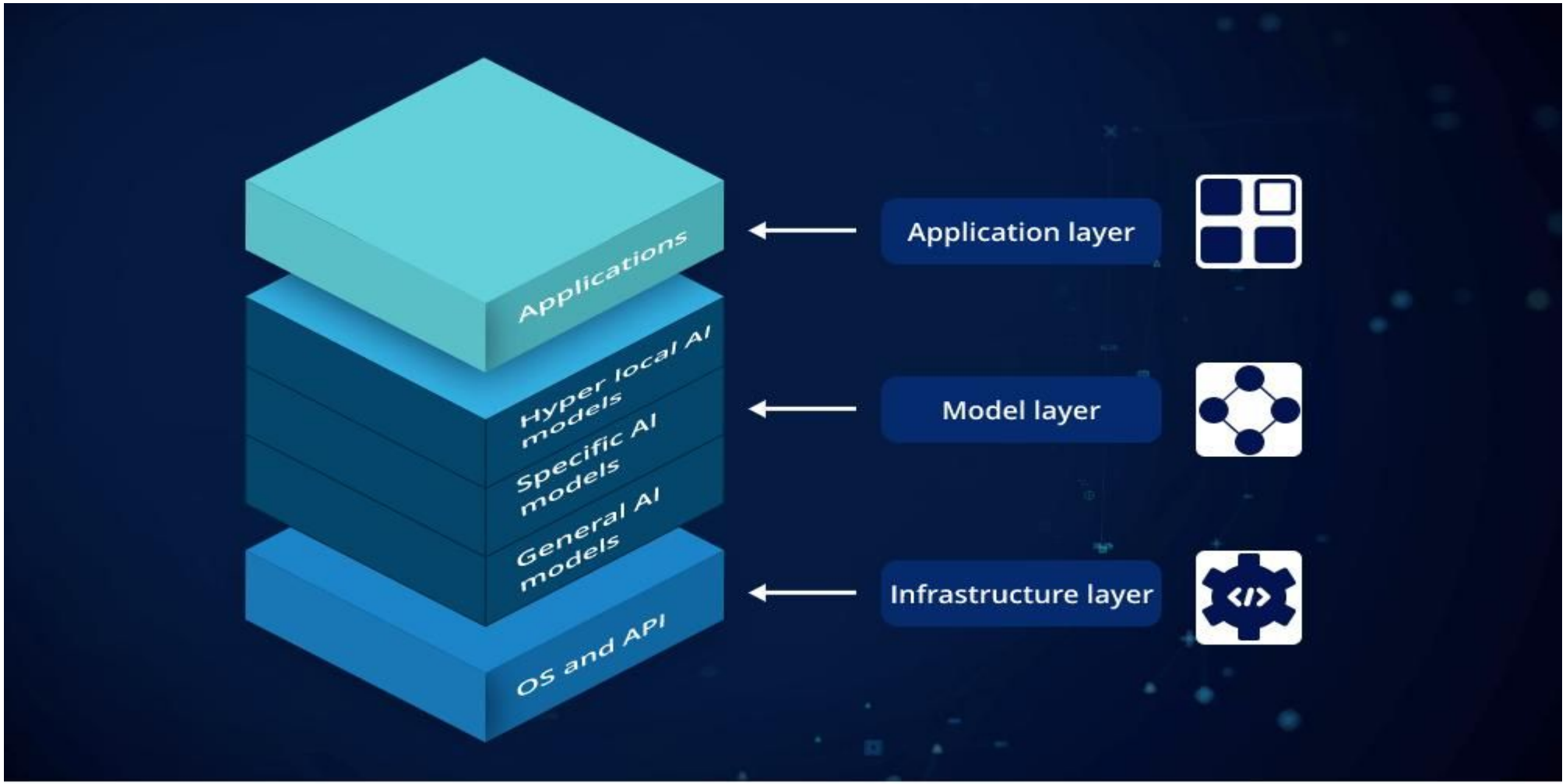
Plans the shortest path to deliver a package (**Deliberative**).

2. Autonomous Drone

Changes direction immediately when it detects an obstacle (**Reactive**).

Plans an efficient flight path to reach its destination (**Deliberative**).

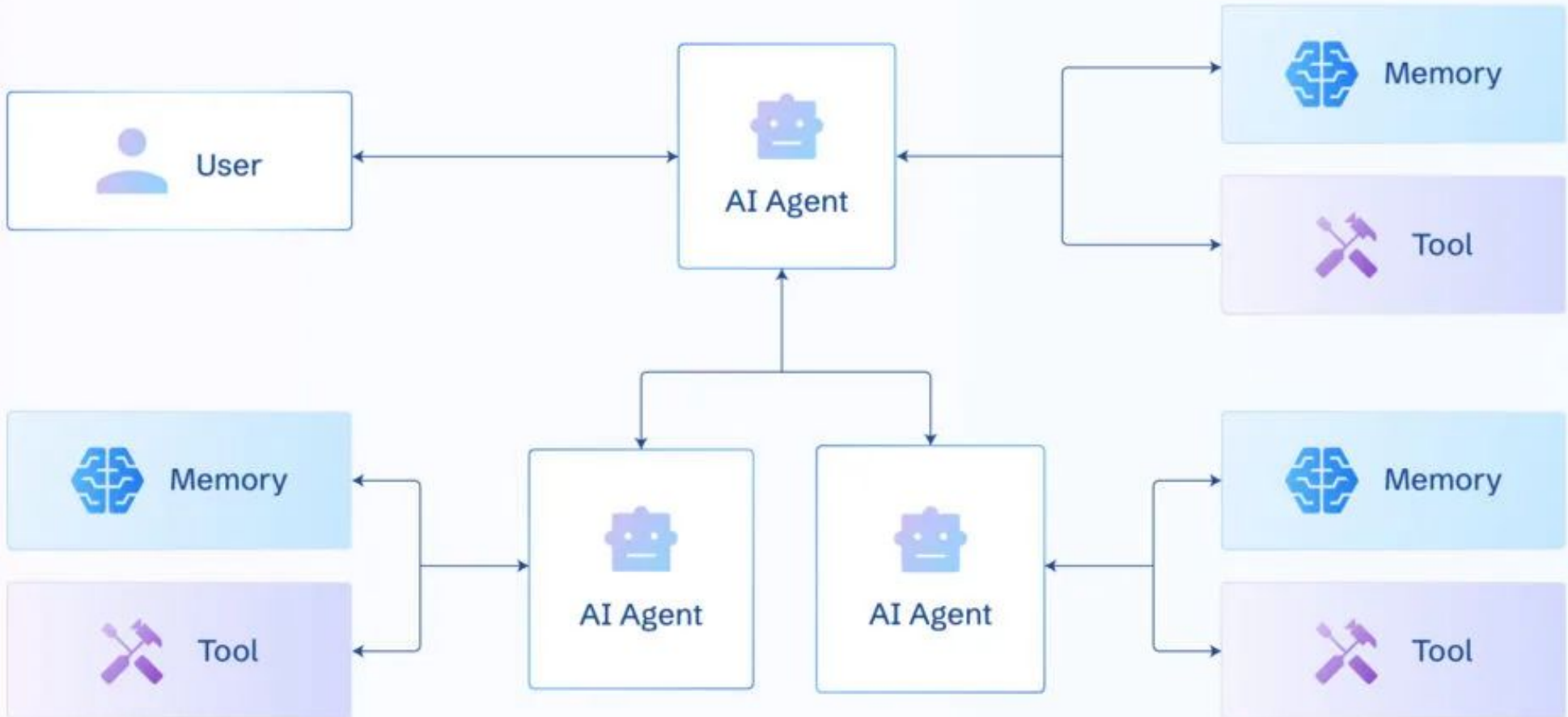
4. Layered Architecture



4. Layered Architecture

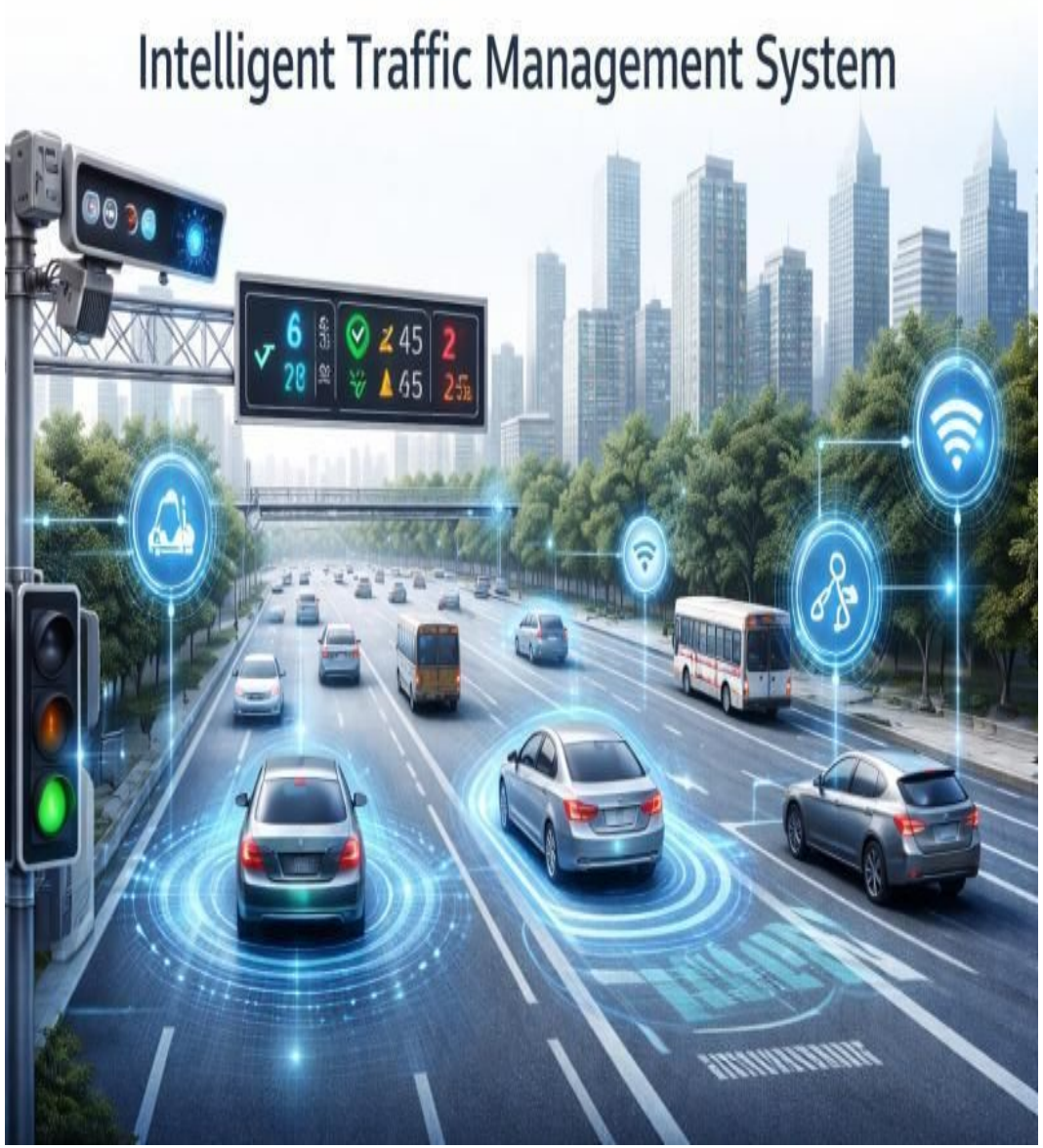
- Layered architecture organizes AI into multiple layers, where each layer performs a specific function.
- **Typical Layers**
  - Reactive Layer
  - Executive (Control) Layer
  - Deliberative (Planning) Layer
- **Example:**
  A **humanoid robot** uses one layer to detect objects, another to plan movements, and another to control its motors.

Multi-Agent Architecture



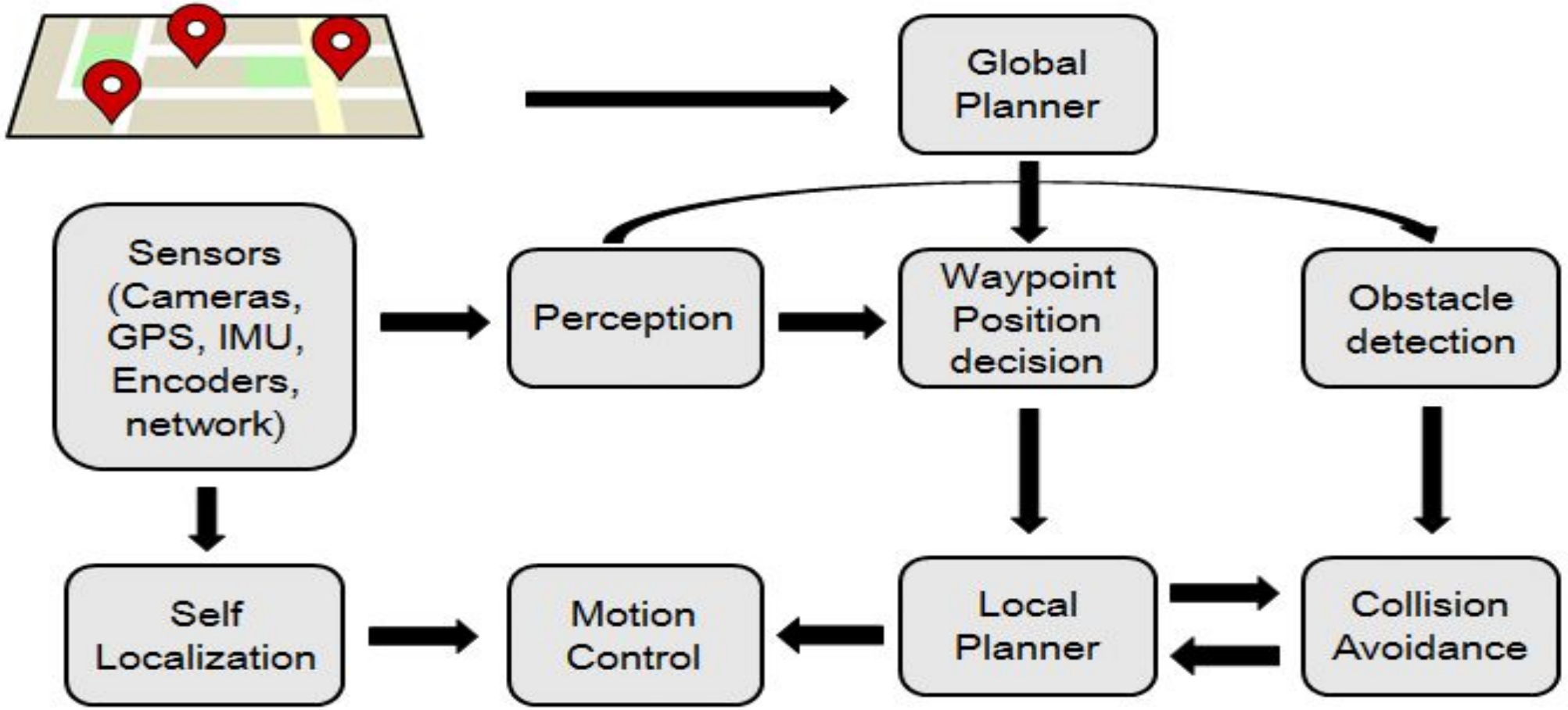
5. **Multi-agent Architectures:**

- Multi-agent systems involve several AI agents, each specialized in handling a specific subset of tasks. This setup allows for more complex operations and dynamic problem-solving. Often, there may still be a lead agent orchestrating the overall process.
- **Strengths:**
  - Better suited for complex and dynamic task environments.
  - Enhanced efficiency through parallel processing capabilities.
  - Utilization of smaller, task-specific models can be more cost-effective and efficient.



- **Smart Traffic Management**
- **Agents:**
  - Traffic signals at different intersections.
- **How it works:**
  - Each signal communicates with nearby signals to adjust green and red light timings based on traffic flow.
- **Benefit:**
  - Reduces traffic congestion and waiting time.

Example:- AI architecture for self driving cars



The architecture shown in the image is a **Hybrid AI Architecture**.

- A **Hybrid Architecture** combines both:
- **Deliberative (Planning) Layer**
  - Thinks and plans before acting.
  - Components in the image:
    - **Global Planner**
    - **Waypoint Position Decision**
    - **Local Planner**
  - These modules calculate the best path and navigation strategy.
- **Reactive (Behavior-Based) Layer**
  - Responds immediately to changes in the environment.
  - Components in the image:
    - **Obstacle Detection**
    - **Collision Avoidance**
  - These modules react instantly to obstacles without waiting for a new global plan.

Intelligent Agents

Agents and Environments

- An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**

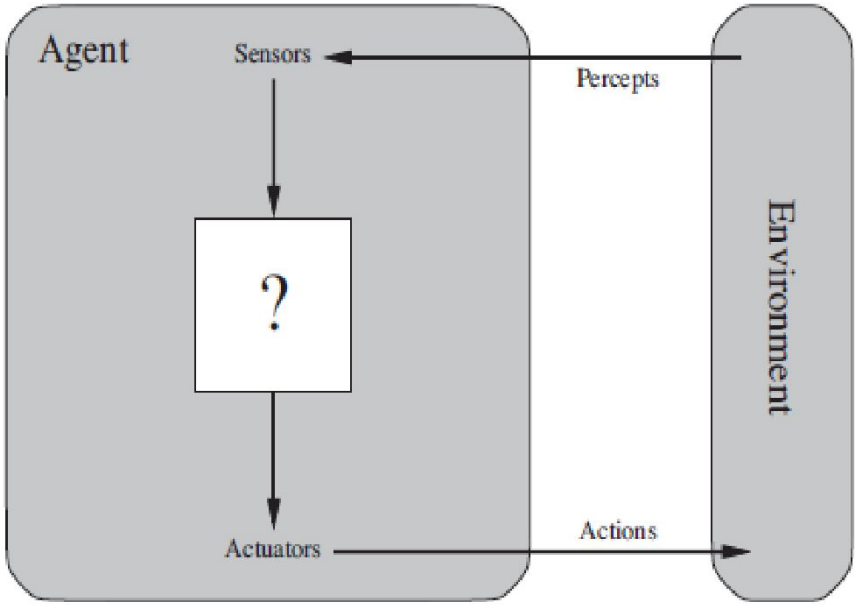


Figure: Agents interact with environments through sensors and actuators.

Agents and Environments

- Eg:
  - A human agent
    - Sensors: eyes, ears, and other organs
    - Actuators: hands, legs, vocal tract, and so on
  - A robotic agent
    - Sensors: cameras and infrared range finders
    - Actuators: various motors
  - A software agent:
    - Sensors: receives keystrokes, file contents, and network packets
    - Actuators: displaying on the screen, writing files, and sending network packets.

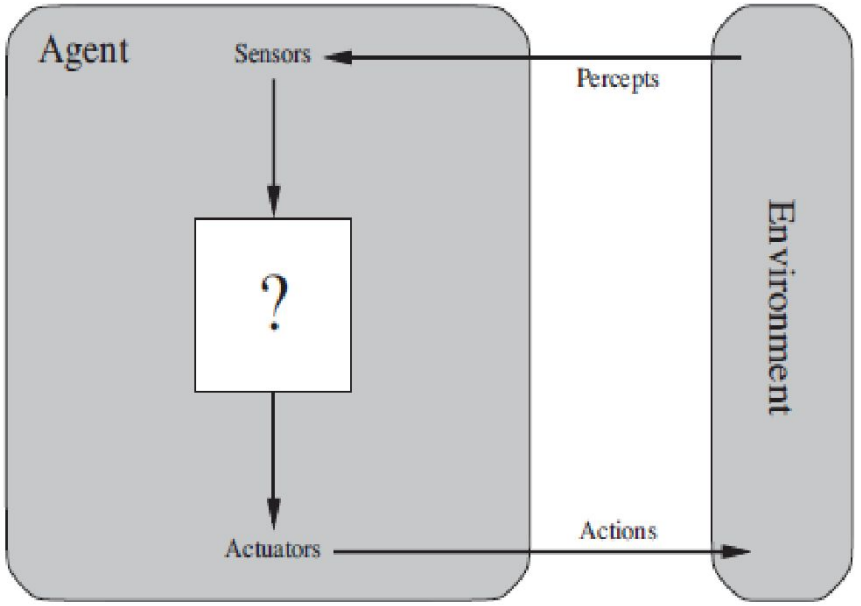


Figure: Agents interact with environments through sensors and actuators.

Agents and Environments

- **Percept** to refer to the agent's perceptual inputs at any given instant.
- **Percept sequence** is the complete history of everything the agent has ever perceived.

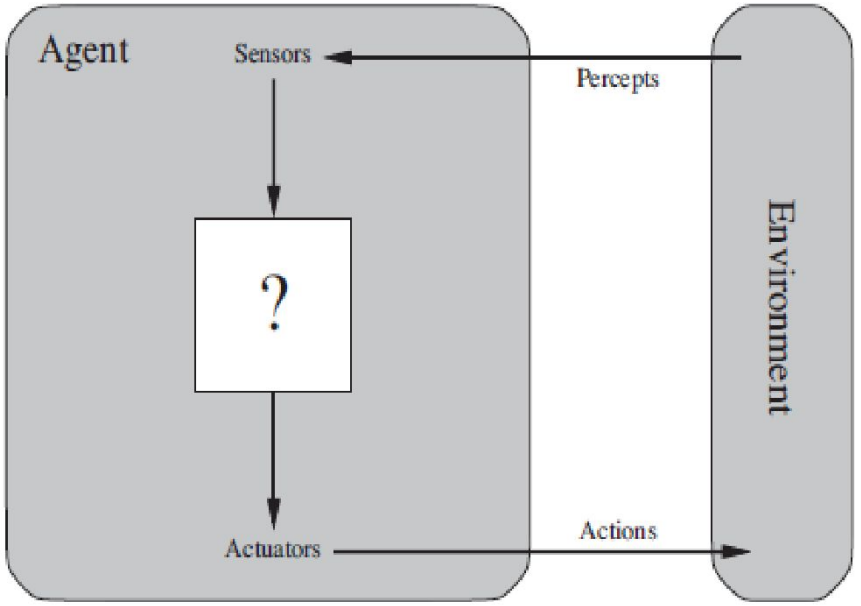


Figure: Agents interact with environments through sensors and actuators.

Agents and Environments

- Eg. Vacuum Cleaner World:
  - Percepts: location and contents e.g. [A, Dirty]
  - Actions: Left, Right, Suck, NoOp

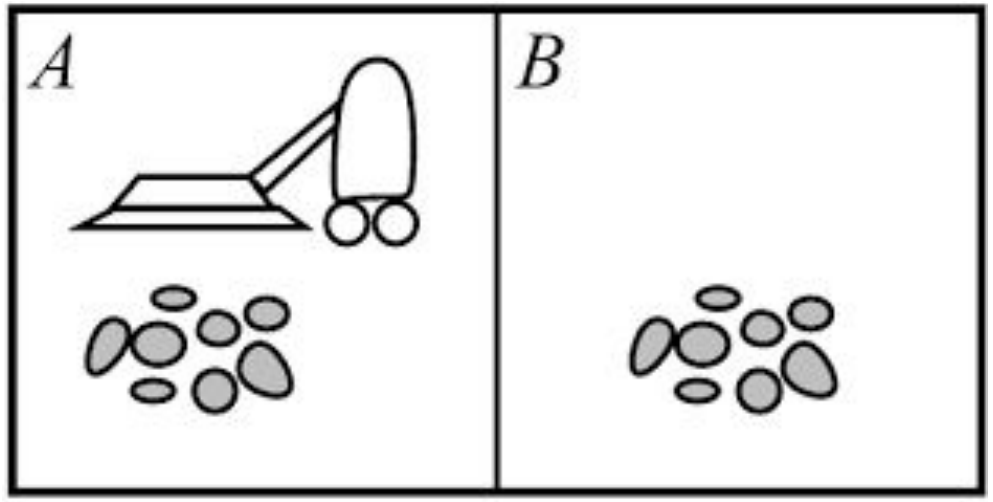


Figure: A vacuum-cleaner world with just two locations.

Agents and Environments

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
...	...
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
...	...

Figure: Partial tabulation of a simple agent function for the vacuum-cleaner world shown in next figure

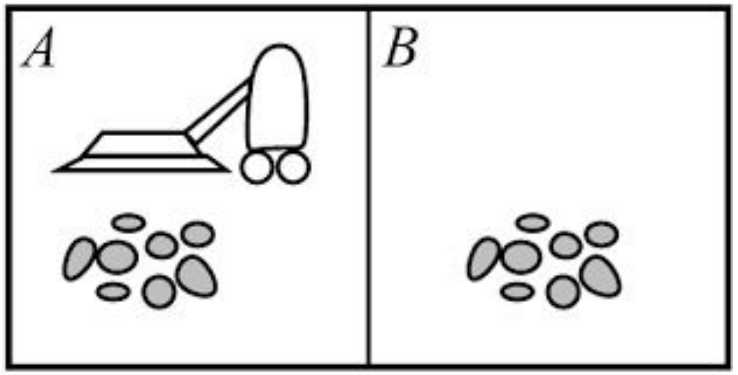


Figure: A vacuum-cleaner world with just two locations.

Good Behavior: Concept of Rationality

- A **rational agent** is one that does the right thing
- **Rationality**
  - What is rational at any given time depends on four things:
    - The performance measure that defines the criterion of success.
    - The agent's prior knowledge of the environment.
    - The actions that the agent can perform.
    - The agent's percept sequence to date.

There are several approaches to defining rationality in AI

- **Goal-Oriented Rationality**:-AI agent is rational if it selects actions that **maximize its expected utility** given its current state and available information.
- **Instrumental Rationality**:-Instrumental rationality refers to an agent's ability to achieve its **specified goals** by means that are both effective and efficient.
- **Bounded Rationality**:-AI systems, particularly in real-world applications, may face **limitations in processing power**, information availability, and time. **Bounded rationality** acknowledges that agents often make decisions that are good enough rather than optimal because they are constrained by these factors.

Good Behavior: Concept of Rationality

- Vacuum Agent PEAS:
  - Performance Measure:
    - minimize energy consumption,
    - maximize dirt pick up.
    - Making this precise: The performance measure awards one point for each clean square at each time step, over a “lifetime” of 1000 time steps
  - Environment:
    - two squares,
    - dirt distribution unknown,
    - assume actions are deterministic and
    - environment is static (clean squares stay clean)
  - Actuators:
    - Left, Right, Suck, NoOp
  - Sensors:
    - agent can perceive its location and
    - whether location is dirty

Good Behavior: Concept of Rationality

- Autonomy:
  - A system is autonomous to the extent that its own behavior is determined by its own experience.
  - Therefore, a system is not autonomous if it is guided by its designer according to a priori decisions.
  - To survive, agents must have:
    - Enough built-in knowledge to survive.
    - The ability to learn.

Nature of Environments

- Automated taxi driving system PEAS:

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure PEAS description of the task environment for an automated taxi.

Nature of Environments

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure Examples of agent types and their PEAS descriptions.

Nature of Environments

- Properties of Environments:
  - Fully Observable/Partially Observable:
    - If an agent's sensors give it access to the complete state of the environment needed to choose an action, the environment is **fully observable**.
    - Such environments are convenient, since the agent is freed from the task of keeping track of the changes in the environment.
  - Deterministic/Stochastic:
    - An environment is **deterministic** if the next state of the environment is completely determined by the current state of the environment and the action of the agent.
    - In a fully observable and deterministic environment, the agent need not deal with uncertainty.

Nature of Environments

- Properties of Environments:
  - Static/Dynamic:
    - A **static** environment does not change while the agent is thinking.
    - The passage of time as an agent deliberates is irrelevant.
    - The agent doesn't need to observe the world during deliberation.
  - Discrete/Continuous.
    - If the number of distinct percepts and actions is limited, the environment is **discrete**, otherwise it is **continuous**.

Nature of Environments

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure Examples of task environments and their characteristics.

Structure of Agents

- Agent program:
  - maps from percept to action and updates internal state.
- ***agent = architecture + program***
- Some agent types:
  - Reflex agents:
    - respond immediately to percepts.
  - Goal-based agents:
    - act in order to achieve their goal(s).
  - Utility-based agents:
    - maximize their own utility function

Structure of Agents

- Simple Vacuum Reflex Agent

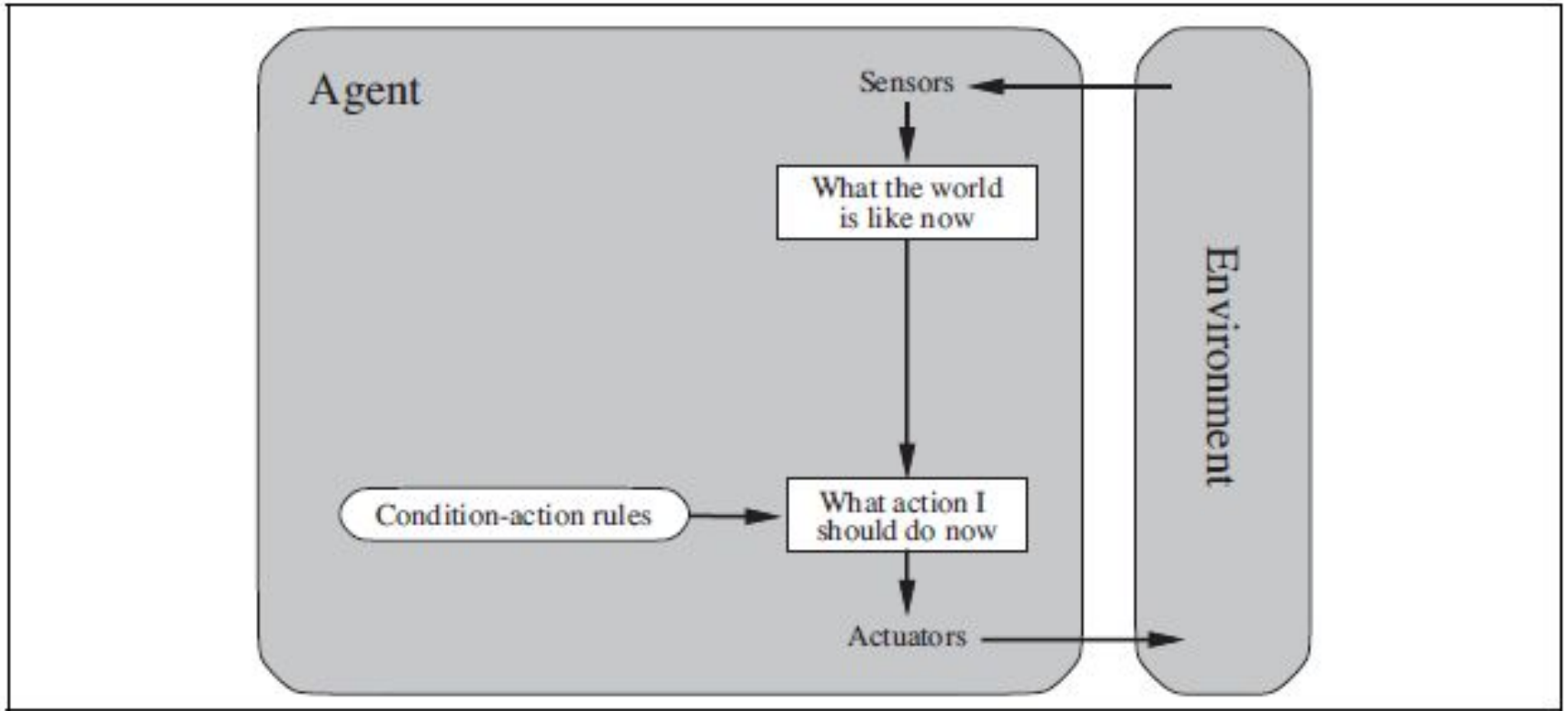


Figure Schematic diagram of a simple reflex agent.

Simple Vacuum Reflex Agent

- A **Simple Vacuum Reflex Agent** is an intelligent agent that decides its action **only based on the current percept (current location and whether it is dirty or clean)**. It does **not** remember past states or plan ahead.
- **Advantages**
  1. Simple and easy to implement.
  2. Fast decision-making.
  3. Works well in fully observable and simple environments.
- **Limitations**
  1. Has **no memory** of previous states.
  2. Cannot handle partially observable environments.
  3. May repeat unnecessary actions (e.g., moving back and forth even when all rooms are clean).
  4. Not suitable for complex real-world problems.

Structure of Agents

- Simple Vacuum Reflex Agent
  - function Vacuum-Agent([location, status]) returns **Action**
    - if status = Dirty then return **Suck**
    - else if location = A then return **Right**
    - else if location = B then return **Left**

Structure of Agents

- **Model-based reflex agents**

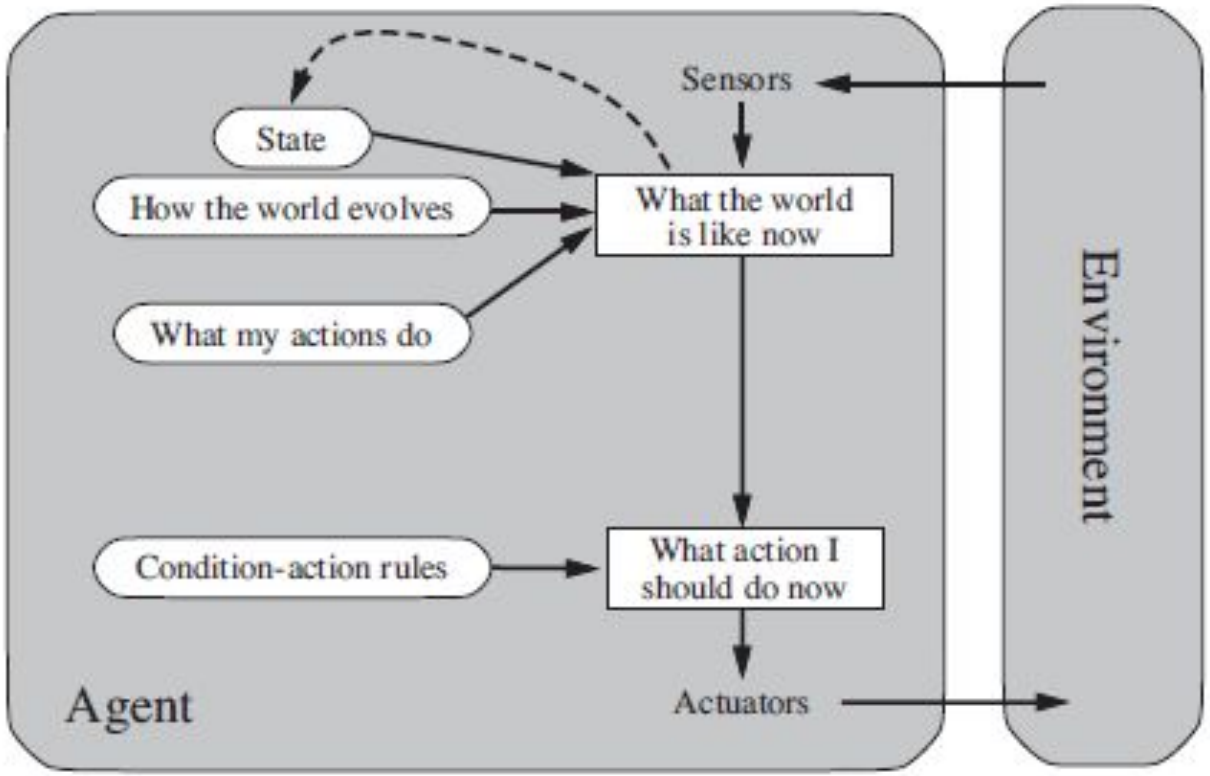


Figure A model-based reflex agent.

Model-Based Reflex Agent

- A **Model-Based Reflex Agent** maintains an **internal state** that represents information about the environment based on previous percepts and actions. It updates this internal state whenever it receives new percepts and then selects an action using condition–action rules.

Components

- **Sensors** – Receive percepts from the environment.
- **Internal State (Memory)** – Stores information about previously observed states.
- **Model of the World** – Describes how the environment changes and how actions affect it.
- **Condition–Action Rules** – Decide the appropriate action based on the internal state.
- **Actuators** – Execute the selected action.

- **Advantages**
  1. Uses memory to store previous observations.
  2. Suitable for **partially observable environments**.
  3. Avoids unnecessary repeated actions.
  4. More intelligent than a simple reflex agent.
- **Limitations**
  1. More complex than a simple reflex agent.
  2. Requires additional memory and computation.
  3. Depends on the accuracy of the internal world model.

- **Real-Life Examples**
  1. **Robot vacuum cleaner (Roomba):** Remembers which rooms have already been cleaned.
  2. **Self-driving car:** Maintains a map of nearby vehicles, pedestrians, and traffic signals, even when some are temporarily out of sensor view.
  3. **Warehouse robot (Amazon):** Remembers shelf locations and previously visited paths to efficiently retrieve items.

- **Key Difference from a Simple Reflex Agent**
  - **Simple Reflex Agent:** "I see dirt → I clean it."
  - **Model-Based Reflex Agent:** "I remember which rooms are clean or dirty, so I can decide where to go next without repeating unnecessary actions."

Structure of Agents

- **Goal-based agents**

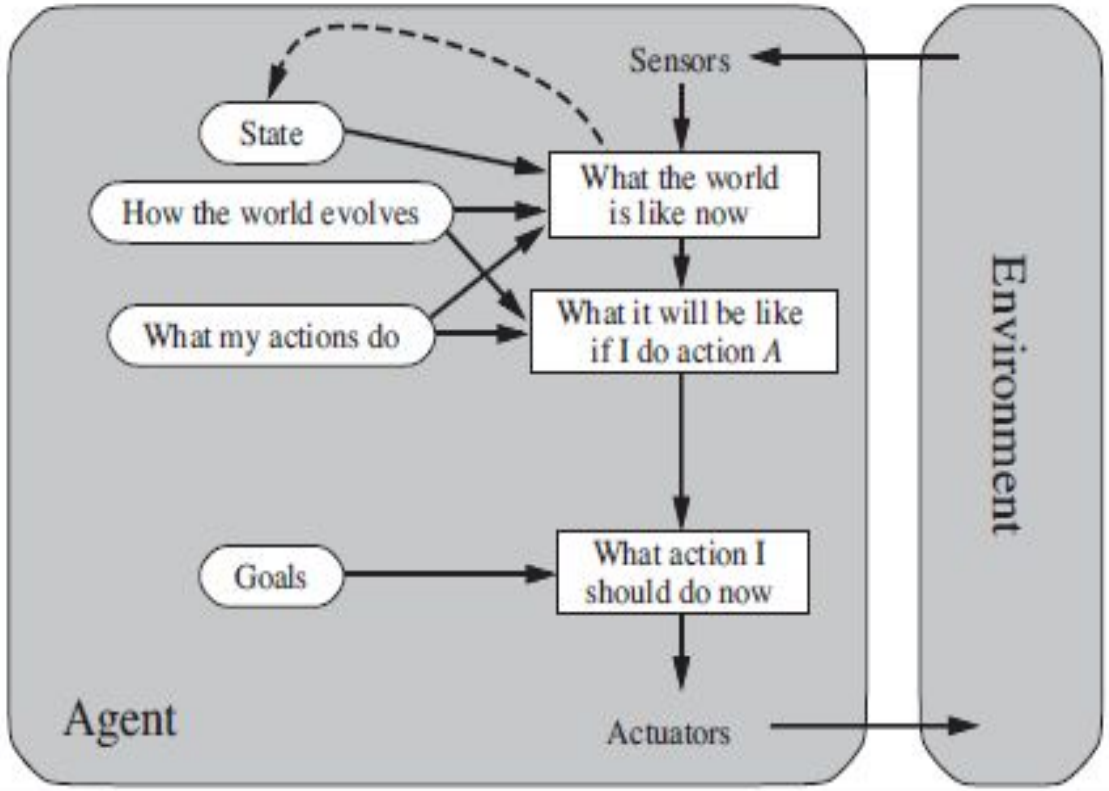


Figure A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

Goal-Based Agent

- A **Goal-Based Agent** is an intelligent agent that selects actions based on a **goal** it wants to achieve. Instead of simply reacting to the current situation, it **plans** a sequence of actions that will lead it from the current state to the desired goal.

Components

- **Sensors** – Perceive the environment.
- **State Information** – Current state of the environment.
- **Goal** – The desired state to be achieved.
- **Search/Planning** – Finds a path from the current state to the goal.
- **Actuators** – Execute the chosen actions.

Characteristics

1. Has a clearly defined **goal**.
2. Plans before taking actions.
3. Can compare different action sequences.
4. Often uses search algorithms such as:
    1. Breadth-First Search (BFS)
    2. Depth-First Search (DFS)
    3. Uniform Cost Search (UCS)
    4. A* Search

- **Advantages**
  1. Can solve complex problems.
  2. Flexible when the environment changes.
  3. Finds a path to achieve a desired goal.
  4. More intelligent than reflex agents.
- **Limitations**
  1. Requires more computation.
  2. Planning can take time.
  3. Needs a clearly defined goal.

- **Real-Life Examples**
  1. **GPS navigation system** – Finds the best route to a destination.
  2. **Warehouse robot** – Plans the shortest path to pick and deliver items.
  3. **Chess-playing AI** – Plans moves to achieve checkmate.
  4. **Drone delivery system** – Plans an optimal route while avoiding obstacles.
- **Key Point**
  - A **Goal-Based Agent** does not simply react to the current situation. It **plans a sequence of actions** to achieve a specified goal, making it more intelligent and adaptable than reflex agents.

Structure of Agents

- **Utility -based agents**

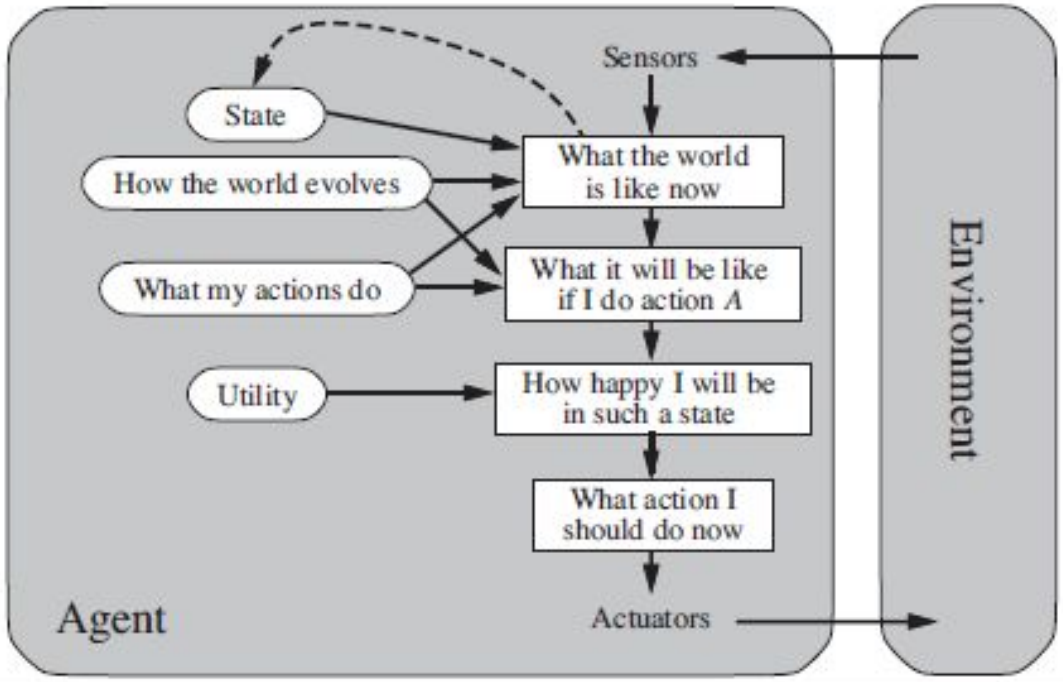


Figure A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

Utility-Based Agent

- A **Utility-Based Agent** is an intelligent agent that selects the **best action** by evaluating how desirable the outcomes are. It uses a **utility function** to assign a numerical value (utility) to each possible outcome and chooses the action with the **highest utility**.
- Unlike a Goal-Based Agent, which only tries to achieve a goal, a Utility-Based Agent tries to achieve the **best possible outcome** among multiple alternatives.

Components

- **Sensors** – Perceive the environment.
- **Current State** – Represents the current situation.
- **Utility Function** – Assigns a utility value to each possible outcome.
- **Decision-Making** – Compares utilities and selects the best action.
- **Actuators** – Execute the chosen action.

Characteristics

1. Makes decisions based on **utility values**.
2. Chooses the **best** option, not just one that reaches the goal.
3. Can handle uncertainty and trade-offs.
4. Considers multiple factors such as cost, time, safety, and efficiency.

- **Advantages**
  1. Chooses the most beneficial action.
  2. Handles multiple possible solutions.
  3. Suitable for uncertain and dynamic environments.
  4. Produces more optimal decisions than goal-based agents.
- **Limitations**
  1. Designing an accurate utility function can be difficult.
  2. Requires more computation than reflex or goal-based agents.
  3. Utility values may be subjective depending on the application.

- **Real-Life Examples**
  1. **GPS Navigation** – Chooses the fastest and safest route.
  2. **Robot Vacuum Cleaner** – Decides whether to clean, recharge, or move based on battery level and dirt.
  3. **E-commerce Recommendation System** – Recommends products with the highest predicted customer satisfaction.
  4. **Self-Driving Car** – Balances safety, speed, fuel efficiency, and traffic conditions.
- **Key Point**
  - A **Utility-Based Agent** not only aims to achieve a goal but also evaluates all possible outcomes and selects the **action with the highest utility**, resulting in the most desirable or optimal solution.

Structure of Agents

- **Learning agents**

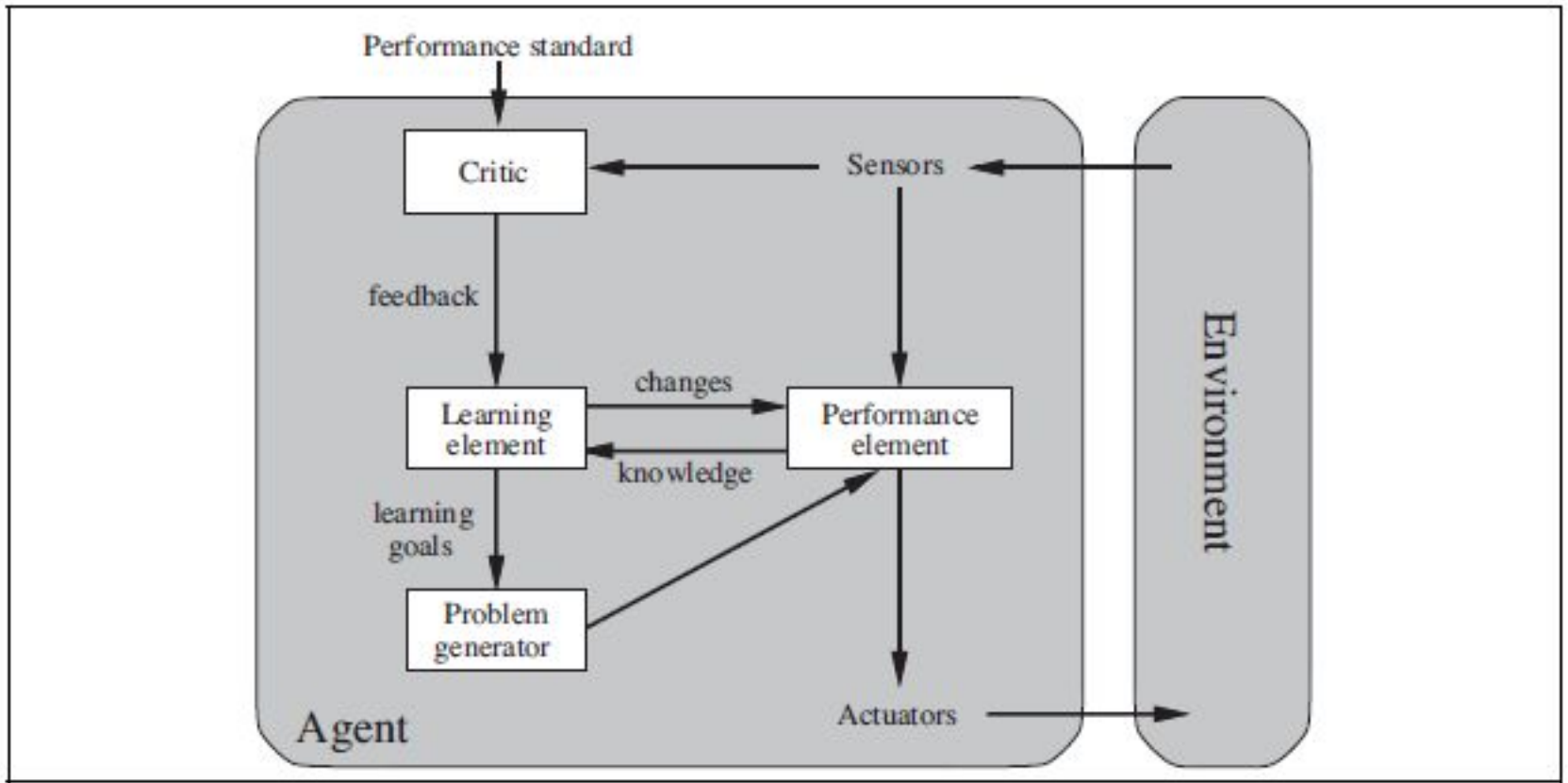


Figure A general learning agent.

Learning Agent

- A **Learning Agent** is an intelligent agent that **learns from its experiences** and improves its performance over time. Unlike other agents, it can adapt to new situations without being explicitly programmed for every scenario.

Components of a Learning Agent

A learning agent has **four main components**:

1. **Performance Element**
    1. Selects actions based on current knowledge.
    2. Interacts with the environment.
2. **Learning Element**
    1. Learns from experience.
    2. Improves the performance element.
3. **Critic**
    1. Evaluates the agent's performance.
    2. Provides feedback (reward or penalty).
4. **Problem Generator**
    1. Suggests exploratory actions to gather new knowledge.
    2. Helps the agent discover better strategies.

Example:- Email Spam Filter

1. Initially, some spam emails are classified incorrectly.
2. The user marks emails as **Spam** or **Not Spam**.
3. The learning agent updates its model using this feedback.
4. Over time, it becomes more accurate at detecting spam.

Characteristics

1. Learns from experience.
2. Improves performance over time.
3. Adapts to changing environments.
4. Uses feedback to modify its behavior.

Advantages

1. Continuously improves performance.
2. Adapts to new situations.
3. Suitable for dynamic and uncertain environments.
4. Reduces the need for manual programming.

Limitations

1. Requires training data or feedback.
2. Learning may take time.
3. Incorrect feedback can lead to poor decisions.
4. Can require significant computational resources.

- **Real-Life Examples**
  1. **Self-driving cars** – Learn from driving data and improve decision-making.
  2. **Email spam filters** – Learn from user feedback to detect spam.
  3. **Netflix and YouTube recommendations** – Learn user preferences and recommend relevant content.
  4. **Amazon product recommendations** – Improve suggestions based on browsing and purchase history.
  5. **Game-playing AI (AlphaGo)** – Learns strategies through practice and feedback.
- **Key Point**
  - A **Learning Agent** is the most advanced type of intelligent agent. It **learns from experience, receives feedback, and continuously improves its performance**, enabling it to adapt to new and changing environments.

Reference

- **Text Books:**
  - Stuart Russell and Peter Norvig, “Artificial Intelligence: A Modern Approach”, Third edition, Pearson, 2003, ISBN :10: 0136042597

Thank You